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Master's Thesis in Robotics, Cognition, Intelligence

Learning Long-Horizon Cyclic Nut Threading with a Dexterous Hand via Hybrid-Teacher Policy Distillation

Lernen langfristiger zyklischer Mutter-Gewindemontage mit einer geschickten Hand durch Policy-Destillation eines hybriden Lehrers

Supervisor Prof. Dr.-Ing. habil. Alois C. Knoll

Advisor Chi Zhang, M.Sc.

Author Carsten Oertel

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Disclaimer

I confirm that this Master's Thesis is my own work and I have documented all sources and material used.

Munich, November 15, 2026

(Carsten Oertel)

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Abstract

Threaded joints are central to assembly because they can be released again and because their threads are standardized. This thesis asks how a robot with a dexterous hand can learn to thread a nut. Threading poses three difficulties at once. The skill has to be learned in simulation and transferred to hardware, although thread contact is hard to simulate. The hand has a high-dimensional action space and many ways to solve the task. Finally, threading is a long-horizon cyclic task: the range of the wrist and the fingers limits each turn, so the hand has to grasp, turn, release, and re-grasp the nut cycle after cycle while errors compound. The thesis decomposes the task during data generation and unifies it at deployment. A privileged reinforcement-learning teacher learns the pinch-grasp turn in a simplified thread simulation, and scripted release and return motions chain its turns into multi-cycle demonstrations. A single visuomotor student is distilled from these demonstrations by conditional flow matching and acts on a metric point cloud from a fixed RGB-D camera and on proprioception. It is trained on simulated data only. A calibrated camera model, visual randomization, and a Cartesian action path shared by teacher, student, and robot carry it zero-shot to a physical dexterous hand-arm system. In simulation, the student completes all ten threading cycles in each of ten trials, with a mean cyclic tracking error of $-1.77°$, a lag of less than two degrees per cycle. On hardware, its zero-shot results on the training nut are TBD, and so is its transfer to nuts of other sizes and profiles.

Kurzfassung

Schraubverbindungen sind zentral für die Montage, weil sie wieder lösbar und ihre Gewinde genormt sind. Diese Arbeit untersucht, wie ein Roboter mit einer geschickten Hand das Aufschrauben einer Mutter lernen kann. Das Aufschrauben stellt drei Schwierigkeiten zugleich. Die Fähigkeit muss in der Simulation gelernt und auf die Hardware übertragen werden, obwohl sich der Gewindekontakt nur schwer simulieren lässt. Die Hand hat einen hochdimensionalen Aktionsraum und viele Möglichkeiten, die Aufgabe zu lösen. Schließlich ist das Aufschrauben eine zyklische Aufgabe mit langem Horizont: Der Bewegungsbereich von Handgelenk und Fingern begrenzt jede Drehung, sodass die Hand die Mutter Zyklus für Zyklus greifen, drehen, loslassen und erneut greifen muss, während sich Fehler aufsummieren. Die Arbeit zerlegt die Aufgabe bei der Datenerzeugung und vereint sie beim Einsatz. Ein privilegierter Lehrer lernt mit Reinforcement Learning die Drehung im Spitzgriff in einer vereinfachten Gewindesimulation, und skriptgesteuerte Loslass- und Rückkehrbewegungen verketten seine Drehungen zu Demonstrationen über mehrere Zyklen. Aus diesen Demonstrationen wird mittels bedingtem Flow Matching eine einzige visuomotorische Schülerpolicy destilliert, die auf einer metrischen Punktwolke einer festen RGB-D-Kamera und auf Propriozeption arbeitet. Sie wird ausschließlich mit simulierten Daten trainiert. Ein kalibriertes Kameramod-

ell, visuelle Randomisierung und ein von Lehrer, Schüler und Roboter geteilter kartesischer Aktionspfad übertragen sie ohne Nachtraining (zero-shot) auf ein reales geschicktes Hand-Arm-System. In der Simulation vollendet die Schülerpolicy in jedem von zehn Versuchen alle zehn Schraubzyklen mit einem mittleren zyklischen Nachführfehler von $-1{,}77°$, also einem Rückstand von weniger als zwei Grad pro Zyklus. Auf der Hardware sind ihre Zero-Shot-Ergebnisse auf der Trainingsmutter TBD, ebenso die Übertragung auf Muttern anderer Größen und Formen.

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List of Abbreviations

ACT	Action Chunking with Transformers
ADR	Automatic Domain Randomization
BC	Behavior Cloning
CFM	Conditional Flow Matching
DAgger	Dataset Aggregation
DoF	Degree of Freedom
DP3	3D Diffusion Policy
DR	Domain Randomization
EMA	Exponential Moving Average
FR3	Franka Research 3
IL	Imitation Learning
LSTM	Long Short-Term Memory
MDP	Markov Decision Process
MLP	Multilayer Perceptron
ODE	Ordinary Differential Equation
OSC	Operational-Space Control
PD	Proportional-Derivative
POMDP	Partially Observable Markov Decision Process
PPO	Proximal Policy Optimization
RGB-D	Color and Depth Image
RL	Reinforcement Learning
RMSE	Root Mean Square Error
SDF	Signed Distance Field
TCP	Tool Center Point

List of Symbols

The following symbols are used with the same meaning in all chapters.

Symbol	Meaning
Task	
$\mathcal{S}$	ordered phase sequence of one threading cycle
$m_t \in \mathcal{M} = \mathcal{M}_{\text{RL}} \cup \mathcal{M}_{\text{script}}$	active phase at step t; learned phases $\mathcal{M}_{\text{RL}}$, scripted phases $\mathcal{M}_{\text{script}}$
K	number of cycles per trial ($K = 10$)
$\Delta\varphi^*$	desired clockwise nut rotation per cycle (60°)
φ_k	unwrapped nut angle after the goal-reaching phase of cycle k
e_k, e_{ep}	cyclic and episodic (accumulated) tracking error
$C(\tau)$, $E(\tau)$	completed cycles and tracking cost of a trajectory τ
ρ	thread pitch
θ_t, d_t	revolute and prismatic coordinate of the simplified thread joint
Teacher	
$\boldsymbol{x}_t$	full simulator state
$\boldsymbol{z}_t^{\pi} = g_{\pi}(\boldsymbol{x}_t, \boldsymbol{\xi}_t)$	actor observation with observation noise $\boldsymbol{\xi}_t$
$\boldsymbol{h}_t$	recurrent hidden state of the actor
$\pi_{\text{T}}(\boldsymbol{a} \mid \boldsymbol{z}^{\pi}, \boldsymbol{h})$, $\bar{\pi}_{\text{T}}$	privileged RL teacher and its deterministic mean
κ_m	scripted controller of transition phase $m \in \mathcal{M}_{\text{script}}$
π_{HT}	hybrid teacher
$\boldsymbol{a}_t \in [-1, 1]^9$	normalized Cartesian-controller action
r_t, $R_{i,t}$, $P_{j,t}$	shaped reward, reward terms, penalty terms
Student	
$\boldsymbol{y}_t = (\boldsymbol{O}_t^{\text{RGBD}}, \boldsymbol{s}_t, p_t, \boldsymbol{e}_t^m)$	student observation: RGB-D image, proprioception, progress, phase encoding
$\mathcal{P}_t$, $\boldsymbol{v}_t \in \mathbb{R}^{64}$	point cloud and visual feature
$\boldsymbol{c}_t \in \mathbb{R}^{256}$	fused condition vector
H	action horizon (chunk length, $H = 8$)
$\boldsymbol{U}_t^D = [\boldsymbol{u}_t, \ldots, \boldsymbol{u}_{t+H-1}]$	demonstrated action chunk
$\boldsymbol{U}_t^0 \sim \mathcal{N}(\boldsymbol{0}, I)$	Gaussian source sample
$\lambda \in [0, 1]$	flow time
$\boldsymbol{U}_t^{\lambda}$	interpolant between source and target chunk
$\boldsymbol{v}_{\theta}(\boldsymbol{U}^{\lambda}, \lambda \mid \boldsymbol{c})$	learned conditional velocity field
N	number of ODE integration steps
β	probability of teacher control during online DAgger
$\mathcal{D}_{\text{sim}}$, $\mathcal{D}$	simulated demonstration set and the training set formed from it
Control	
$\boldsymbol{q}$, $\dot{\boldsymbol{q}}$	joint positions and velocities
$\boldsymbol{\tau}$	joint torques
J, M, Λ	geometric Jacobian, joint-space inertia, task-space inertia
K_p, K_d	proportional and derivative gains

Chapter 1

Introduction

1.1 Motivation

Threaded fasteners such as a nut on a bolt join parts so that the joint can be released again, which makes them the standard choice wherever products are assembled in the field, maintained, adjusted, or taken apart [1], [2]. They appear in almost every industry. The halves of a cast engine housing are bolted together, airplane seats are mounted with threaded bolts, and the injector of a rocket engine is bolted onto the combustion chamber. Many valve and pipe connections consist of two threaded halves that are screwed together with a sealing material, so that the liquid stays inside, from the plumbing of a house to the water cooling loop of a gaming PC. These applications have the same needs. The product has to be manufacturable in modules, assembled in a defined process, and opened again without damage for inspection, maintenance, repair, and development.

The vast majority of people have also met threaded fasteners in everyday life, for example when assembling IKEA furniture or fixing household objects and appliances. For a human, the process is intuitive. You pick up a screw, find the hole, set the tip of a Philips screwdriver into the screw head, and turn. If the screw does not bite, you back off, straighten it in the hole, and turn again. Industry solves the same task differently. In industrial production, turning the fastener along the thread is done by dedicated equipment, as Figure 1.1 illustrates. Handheld and stationary nutrunners drive the fastener with a powered spindle, multi-spindle stations tighten several fasteners of one part at once, and robots carry screwdriving end effectors that feed and drive fasteners. None of these machines has to find the hole. A fixture clamps the part in a known pose, so the hole is always at the same place. The screw or nut arrives from a feeder in a known orientation, and the tool is matched to that one fastener type. This setup pays off in high-volume production. It stops working where parts vary, where the cell is not built around one product, and where batches are small. It also stops working in maintenance and disassembly, where the hole is wherever the part happens to be, as it is for a person kneeling next to a half-built wardrobe.

People handle the cases that dedicated equipment cannot. They pick up a nut, place it on a bolt, and turn it with their fingers, and they use the same hands to operate a wrench or a screwdriver. A robot with a dexterous, multi-fingered hand could in principle do the same. It could work with the parts and tools that were designed for people, in workcells that were designed for people, without a dedicated tool for every fastener [4], [5]. Learning-based control has made in-hand manipulation with such hands practical in simulation and on hardware [5], [6], [7]. Robots have fastened screws with manual screwdrivers [8], [9], and a single learned policy has manipulated a range of everyday tools with a dexterous hand [10]. The vision behind this thesis is a dexterous robot that helps in manufacturing and assembly in a world built for humans. Threading a nut by hand is one of the basic skills such a robot



(a)

(b) Placeholder: multi-spindle station

(c) Placeholder: robot with screwdriving end effector

(d) Placeholder: fixture presenting the part

Figure 1.1: How threading is automated today: (a) a stationary nutrunner (image from [3]), (b) a multi-spindle station that tightens several fasteners of one part at once, (c) a robot with a screwdriving end effector, and (d) a fixture that presents the part in a fixed pose. Each solution is built for one fastener type and a known part pose.

needs. For humans the skill is trivial, but it is not for robots. This is an instance of Moravec's paradox: it is comparatively easy to make computers perform well on tasks that people find intellectually demanding, and difficult to give them the perception and mobility skills of a small child [11]. Industrial fastening shows both sides of this paradox. A nutrunner on a fixture drives a fastener without difficulty, because the cell design has already solved the hard part: the part is held in a known pose and the fastener is presented in a known orientation. Without this structure, a robot has to find the hole, align the fastener, and engage the thread, which a person does without thinking. If a dexterous robot is to thread, it has to learn the skill, and the next section explains why that is hard.

1.2 Problem: How Can a Robot Learn Threading?

Teaching a robot to thread a nut with a dexterous hand raises three difficulties at once. First, the skill has to be learned in simulation and transferred to hardware, although thread contact is very hard to simulate. Second, the dexterous hand has a high-dimensional action space and countless ways to solve the task. Third, threading is a long-horizon cyclic task. The following subsections treat the three difficulties in turn, and each ends with the requirement it puts on the method.

1.2.1 Simulation

Learning a contact-rich skill directly on hardware is slow and unsafe. Reinforcement learning (RL) needs many trials, every trial costs robot time and wear, and exploration produces collisions and excessive forces [12]. Simulation removes these limits. GPU-parallel simulators run thousands of environments at once and have made RL feasible for dexterous manipulation and for contact-rich assembly [7], [13], [14].

Thread contact, however, is among the hardest interactions to simulate. Engaged threads produce thousands of simultaneous contacts [14], and simulating them faithfully needs dedicated contact methods [14] that remain expensive at the scale of parallel rollouts that RL needs.

The GPU-parallel engines that make this scale possible, such as PhysX and MuJoCo, model contact through interpenetration. When two bodies overlap, the solver measures the penetration depth and generates a repulsive force that pushes the bodies apart again (Figure 1.2). Contact is therefore never strictly rigid. It tolerates a small overlap, and the repulsive force grows with that overlap.

For large bodies this tolerance is harmless. For threads it is not, because the thread

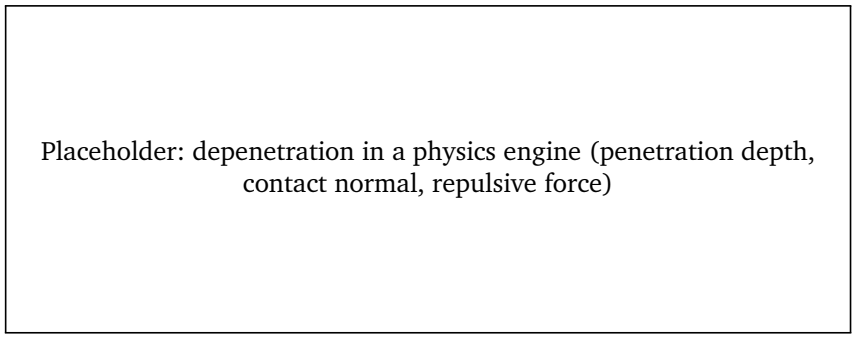


Figure 1.2: How a physics engine resolves contact. Two bodies that overlap by a penetration depth d receive a repulsive force along the contact normal that grows with d and pushes them apart.

geometry is small. The pitch is 3 to 4 mm for the nuts used in this thesis, and the flanks and crests are smaller still. At this scale, the overlap that the solver tolerates is no longer negligible against the geometry. A nut that is pushed along the axis can sink into the opposing flank and slide across the crest into the next groove, instead of following the helix. The repulsive force does not forbid this path, it only makes it more expensive. In simulation, the nut can thus phase through the threads.

A learning policy finds such shortcuts. If the reward pays for axial progress of the nut, the policy can collect it by pushing the nut through the threads rather than by turning it along the helix (Figure 1.3). The resulting behavior solves the simulated task and fails on hardware, where the threads are rigid and the nut advances only by rotating. Faithful thread contact is therefore both expensive to simulate and easy to exploit, and any simplification of the thread opens a gap in contact between simulation and reality. A deployed policy also has to perceive the nut through a camera, and rendered images differ from real ones, which opens a second gap in perception. Collecting real data to close these gaps would bring back the cost that simulation was meant to avoid.

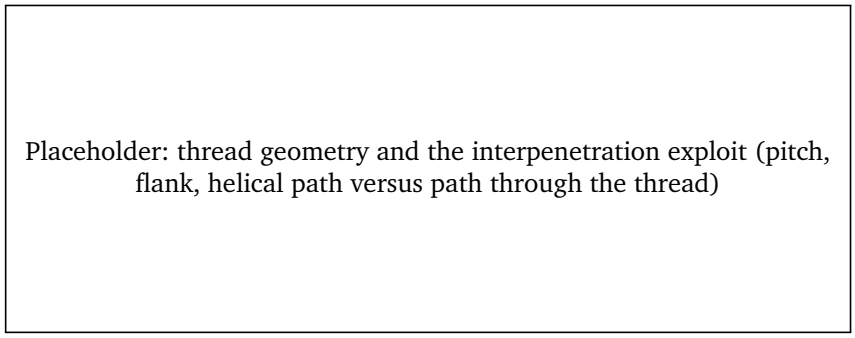


Figure 1.3: How interpenetration can be exploited in thread simulation. A reward for axial progress of the nut is collected on the helical path by turning the nut, but also on a shortcut in which the nut interpenetrates the thread flank and crosses the crest into the next groove.

Requirement: a thread model that is cheap enough for massively parallel RL and cannot be exploited through interpenetration, and a transfer recipe that carries the learned skill to hardware without real-world data.

1.2.2 Threading with a Dexterous Hand

A multi-fingered hand on an arm has far more actuated joints than a parallel-jaw gripper, and its action space is correspondingly high-dimensional [6], [15]. The task also has a large

solution space. People thread a nut with a single finger, with a pinch between thumb and index finger, or with a grasp of several fingers, depending on how strongly the thread resists. A policy that may choose among all of these strategies has to explore a vast space of finger motions, and a reward that steers it toward one reliable strategy is hard to design [12].

A dexterous hand also has to acquire the grasp, keep it under load while the nut turns, and release it again. Learned assembly with parallel-jaw grippers does not face this part of the problem. There, the threading or insertion policy starts every episode with the part already held, so it learns to move a part that is grasped, and grasping, holding, and re-grasping are not part of what it learns [14], [16], [17]. Learned dexterous threading has so far addressed the spinning of an already engaged fastener [18] and goal-conditioned screwing of furniture parts [19].

Requirement: a learned policy that controls the hand and the arm together, with the solution space narrowed to one grasp strategy, the pinch grasp, which the policy has to acquire, hold, and release.

1.2.3 Threading as a Long-Horizon Cyclic Task

A long-horizon cyclic task repeats a fixed sequence of phases many times while the task state advances from one repetition to the next. Threading with a pinch grasp is such a task. One turn is limited by the range of the wrist and the fingers, so the hand has to approach the nut, grasp it, turn it, release it, and return for the next grasp, cycle after cycle. Meanwhile the nut advances along the thread by a fraction of a pitch in every cycle, so no two cycles start from the same state. Many tasks in assembly and everyday manipulation share this structure, for example driving a screw, turning a crank, winding a cable, or wiping a surface.

Three properties make such tasks hard. First, errors compound across cycles. A grasp that lands slightly off the intended faces, a release that disturbs the nut, or a return that brings the fingers back at the wrong height carries over into the next cycle and accumulates. For a learned policy this is worse, because small deviations lead it to states it was not trained on, where its errors grow further [20]. Second, failures concentrate at the transitions between phases, where the contact between hand and nut is made or broken, rather than within a single phase. Third, a single RL policy for the whole task needs a reward that covers every phase and every transition, and it has to assign the outcome of a late cycle to decisions made many steps earlier. Reward shaping over many phases and credit assignment over the full horizon break down for such a policy [12].

The common answer is to decompose the task into skills, train one policy per skill, and chain the policies at deployment [15]. Chaining moves the difficulty to the transitions. The final state of one policy has to lie in the region where the next policy succeeds, which needs extra machinery that fine-tunes the sub-policies for their neighbors [15], and several policies together with a switching logic have to be deployed. A second answer is to imitate complete long-horizon demonstrations with one generative policy [21], [22]. This avoids the reward problem, but it needs demonstrations of the whole task, and teleoperated demonstrations of forceful dexterous manipulation are hard to collect [10].

Requirement: a training signal that is easy to specify phase by phase, and one deployable policy that sustains many cycles without switching between separate policies.

Taken together, the three requirements point to one approach: decompose the task where the training signal is generated, and unify it where the policy is deployed.

1.3 Approach

This thesis decomposes the task during data generation and unifies it at deployment, as Figure 1.4 summarizes. Each part of the approach answers one of the difficulties of Section 1.2. A simplified kinematic thread model couples the axial travel of the nut to its rotation while keeping the finger–nut contacts, which makes massively parallel training with proximal policy optimization (PPO) feasible (Section 1.2.1). A privileged RL teacher, which observes the true nut pose in simulation, learns the pinch-grasp turn: approach, grasp, rotation, and goal reaching (Section 1.2.2). Scripted controllers release the nut and return the hand. Chained with the teacher, they form a hybrid teacher that generates complete multi-cycle demonstrations whose phase boundaries are triggered by physical events. A single visuomotor student is then distilled from these demonstrations by conditional flow matching (CFM) [23]. This teacher–student structure follows earlier work that distills privileged policies into sensor-based students [24], [25], [26]. The student cannot observe the true nut pose, so it replaces the teacher's privileged nut state with perception. It acts from proprioception and a fixed RGB-D camera whose model is calibrated against the physical cell. The depth image is converted into a metric point cloud [27], and the rendering is randomized during data generation [24], so that the student sees similar inputs in simulation and on hardware. Teacher, student, and robot share one Cartesian action path. The student therefore runs the whole cycle on hardware without privileged state, without switching between policies, and without real-world data.

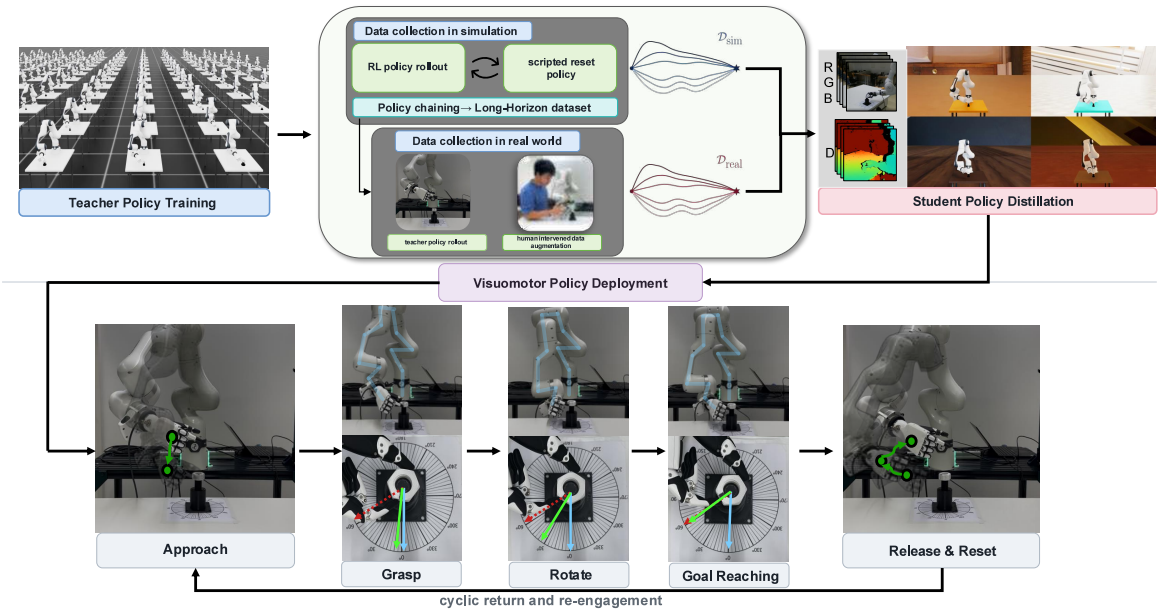


Figure 1.4: Overview of the approach: decompose during data generation, unify at deployment. Top: a privileged RL teacher is trained in parallel simulated workcells; during data collection its rollouts are chained with scripted release-and-reset controllers into long-horizon demonstrations $\mathcal{D}_{\text{sim}}$; a conditional flow-matching student is distilled from them and acts from RGB and depth. Bottom: the deployed student executes approach, grasp, rotation, goal reaching, and release-and-reset on the physical robot and returns for the next cycle.

1.4 Research Questions

Note: Perhaps this section should be removed from the introduction.

The approach rests on two claims, that one student can replace a chain of policies and that it can be trained on simulated data alone, and on one design choice, the student that performs the unification. Three research questions test them.

RQ1 (long horizon). Does one student distilled from complete multi-cycle demonstrations sustain consecutive threading cycles better than a single-cycle policy that is invoked repeatedly?

RQ2 (student design). Which student design distills the hybrid-teacher demonstrations into a policy that sustains the cyclic task? The thesis compares a flow-matching action head with deterministic regression on long short-term memory (LSTM) and Transformer backbones, and a point-cloud encoder with an image-space encoder.

RQ3 (sim-to-real). Can a student trained on simulated demonstrations only transfer zero-shot to the physical hand–arm system?

RQ1 tests the answer to the long-horizon difficulty, RQ2 asks which student makes the unification work, and RQ3 tests the answer to the transfer difficulty. Chapter 4 answers RQ1 and RQ2 in simulation, and Chapter 5 answers RQ3 on hardware. The answer to the dexterity difficulty, the privileged teacher, is tested by the teacher training that Chapter 4 reports before the student experiments.

1.5 Contributions

The thesis makes the following contributions.

1. A formulation of cyclic nut threading as an ordered sequence of phases with explicit cyclic and episodic tracking errors, and a hybrid teacher that combines a privileged RL policy for approach, grasp, rotation, and goal reaching with scripted controllers for release and return. The hybrid teacher generates coherent multi-cycle demonstrations with physically gated phase boundaries.

2. A unified visuomotor student distilled by conditional flow matching from a point-cloud representation of an aligned RGB-D view, proprioception, a global trajectory-progress scalar, and a process-phase encoding. A systematic ablation compares it with deterministic regression heads, alternative vision encoders, online dataset aggregation (DAgger) variants, privileged observations, and a joint-space action contract.

3. A zero-shot sim-to-real recipe for the student: a camera model calibrated against the physical cell, a metric point cloud cropped to the workspace, visual randomization of the rendered demonstrations, and a Cartesian action path shared by teacher, student, and robot.

4. An evaluation of the student in simulation on the hexagonal training nut and on hardware across three nut sizes and four nut profiles, three of which were not seen in training. The simulation results are reported in Chapter 4; the hardware results are TBD.

The work continues the author's semester thesis [26], which distilled a privileged grasping teacher into an RGB student with DAgger on a dexterous hand–arm system. It was carried out together with the advisor, who developed substantial parts of the hybrid teacher, the distillation stack, and the hardware experiments; the acknowledgments state this attribution.

1.6 Outline

This thesis comprises seven chapters: Introduction, State of the Art, Methodology, Experiments in Simulation, Experiments on Real Hardware, Discussion, and Conclusion and Future Work. This chapter motivated threading as a skill for dexterous robots, named the difficulties of learning it, and stated the approach and the research questions. Chapter 2 reviews the background and related work, Chapter 3 presents the method, Chapters 4 and 5 report the experiments in simulation and on hardware, Chapter 6 interprets the results, and Chapter 7 concludes the thesis. The remainder of this section outlines these chapters in more detail.

Chapter 2 provides the background that the method builds on: the thread geometry that the thread model uses, RL with privileged information, imitation learning and compounding error, generative action heads, visual representations for manipulation, and operational-space control. It then reviews related work on robotic assembly and threading, dexterous manipulation, sim-to-real transfer, long-horizon tasks and policy chaining, and teacher–student distillation.

Chapter 3 presents the method, with each part answering one difficulty of Section 1.2: the problem formulation, the simulated system with its thread model and controller, the privileged teacher, the hybrid teacher and the generation of demonstrations, the distillation of the student by conditional flow matching, and the evaluation protocol.

Chapter 4 reports the experiments in simulation. It covers teacher training, the main ablations that answer RQ1 and RQ2, the design ablations of the student, and the comparison of offline and online distillation.

Chapter 5 describes the hardware setup and the zero-shot transfer protocol and reports the real-world evaluation across nut sizes and geometries, which answers RQ3.

Chapter 6 interprets the results with respect to the research questions, discusses what they mean for robotic assembly, and states the limitations of the work.

Chapter 7 returns to the assembly motivation of this chapter, summarizes what a dexterous robot can now do in threading and what is still missing, and outlines future work.

Chapter 2

State of the Art

This chapter provides the background that the method of Chapter 3 builds on and places the thesis within related work. Section 2.1 introduces the theoretical concepts in the order in which the method uses them: the geometry of threaded joints that the thread model reproduces, reinforcement learning with privileged critics for the teacher, imitation learning and its compounding-error problem for the distillation, generative action heads and visual representations for the student, and operational-space control for the action path that teacher, student, and robot share. Section 2.2 reviews related work along the three difficulties that threading poses for a robot: it covers robotic assembly and threading, dexterous manipulation with learned policies, sim-to-real transfer for contact-rich manipulation, long-horizon tasks and policy chaining, and teacher–student distillation, and it closes with the gap that this thesis addresses.

2.1 Theoretical Background

2.1.1 Threaded Joints and Thread Geometry

Screws are divided into fastening screws and motion screws, and a nut on a bolt is a fastening joint [2]. Threaded fasteners make a joint nonpermanent, so that it can be released again for field assembly, disassembly, maintenance, and adjustment [1]. Their threads are standardized, for example in the ISO metric M series [1], whose geometry Figure 2.1 shows. The thread converts a large circumferential motion into a small axial motion, and a small circumferential force into a large axial force [2]. Assembling a threaded joint and securing it are treated as separate design topics [2]. This thesis addresses the assembly motion, turning the nut along the thread, and not the securing of the joint.

The quantity that links rotation and axial travel is the pitch ρ, the distance between adjacent threads [1]. For a single-start thread, adjacent threads are successive turns of one helix. A nut that turns by an angle $\Delta\theta$ about the bolt axis therefore advances along the axis by $\rho\,\Delta\theta/(2\pi)$, that is, by one pitch per revolution. Because a partial turn advances the nut by only a fraction of a pitch, the axial position at which a hand must grasp the nut changes with every turn. The simplified thread model of Section 3.2.2 keeps exactly this relation: it replaces the flank contacts by a kinematic coupling that enforces one pitch of advance per revolution and represents the resistance of the thread to turning by a friction torque.

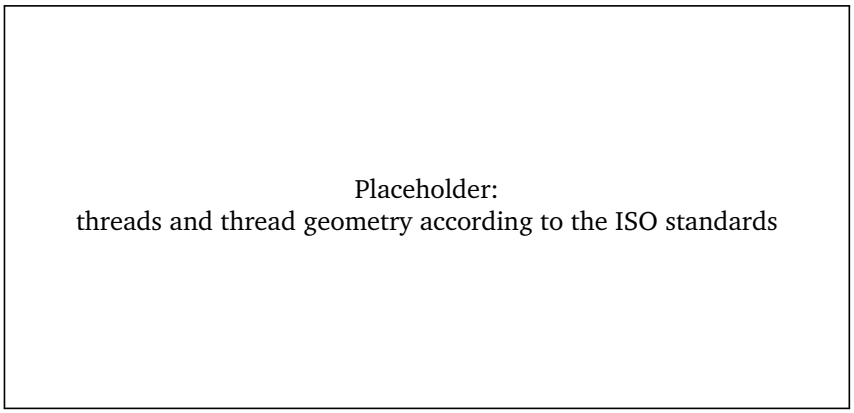


Figure 2.1: Threads and thread geometry according to the ISO standards. Placeholder.

2.1.2 Reinforcement Learning with Privileged Information

Reinforcement learning formalizes sequential decision making as a Markov decision process (MDP) $(\mathcal{X}, \mathcal{A}, P, r, \gamma)$ with state space $\mathcal{X}$, action space $\mathcal{A}$, transition kernel $P(\boldsymbol{x}_{t+1} \mid \boldsymbol{x}_t, \boldsymbol{a}_t)$, reward $r(\boldsymbol{x}_t, \boldsymbol{a}_t)$, and discount factor $\gamma \in [0, 1)$ [12]. A policy $\pi(\boldsymbol{a} \mid \boldsymbol{x})$ is sought that maximizes the expected discounted return $J(\pi) = \mathbb{E}_\pi[\sum_t \gamma^t r_t]$. In robot manipulation the full state is not observable at deployment. The problem is then a partially observable Markov decision process (POMDP), in which the policy acts on observations $\boldsymbol{z}_t \sim O(\cdot \mid \boldsymbol{x}_t)$ instead of states.

PPO [28] is the on-policy actor–critic algorithm used throughout this thesis. The actor π_θ is updated by maximizing the clipped surrogate objective

$$L^{\mathrm{PPO}}(\theta) = \mathbb{E}_t\Big[\min\big(\eta_t(\theta)\hat{A}_t,\ \mathrm{clip}(\eta_t(\theta), 1-\epsilon, 1+\epsilon)\hat{A}_t\big)\Big], \qquad \eta_t(\theta) = \frac{\pi_\theta(\boldsymbol{a}_t \mid \boldsymbol{z}_t)}{\pi_{\theta_{\mathrm{old}}}(\boldsymbol{a}_t \mid \boldsymbol{z}_t)}, \tag{2.1}$$

where $\eta_t(\theta)$ is the probability ratio between the new and the old policy, ϵ is the clipping parameter, and $\hat{A}_t$ is an advantage estimate computed from the critic's value prediction. The clipping bounds the policy change per update and makes PPO stable on the thousands of parallel environments that GPU simulators provide.

In an asymmetric actor–critic the actor and the critic receive different inputs [29]. The critic estimates the value function from the complete simulator state, including quantities that no sensor measures. The actor observes only a noisy subset that can later be reproduced on hardware or distilled into a sensor-based policy. Because the critic is discarded after training, the privileged information shapes the gradients without entering the deployed policy. When the actor observation is itself a noisy or delayed function of the state, a recurrent actor with a LSTM cell [30] integrates the observation history into a hidden state $\boldsymbol{h}_t$ and thereby recovers part of the information lost to partial observability. This recipe, together with domain randomization of physical and controller parameters, underlies the sim-to-real results in dexterous manipulation reported in [7], [13], [24], [25], and it is the recipe of the privileged teacher in Section 3.3.

2.1.3 Imitation Learning and Compounding Error

Imitation learning (IL) trains a policy from demonstrations rather than from a reward. Given a dataset $\mathcal{D} = \{(\boldsymbol{y}_t, \boldsymbol{u}_t^*)\}$ of observations and expert actions, behavior cloning (BC) fits a policy by supervised regression,

$$\theta^* = \arg\min_\theta \mathbb{E}_{(\boldsymbol{y}, \boldsymbol{u}^*) \sim \mathcal{D}}[\ell(\pi_\theta(\boldsymbol{y}), \boldsymbol{u}^*)], \tag{2.2}$$

with a per-sample loss ℓ such as the squared error. BC is simple and data-efficient, but it is trained on the state distribution induced by the expert and evaluated on the state distribution induced by the student. Small prediction errors move the student into states the expert never visited, where its predictions are unreliable, and the error grows with the horizon. Ross et al. [20] show that the expected cost of a BC policy with per-step error ε can grow as $O(T^2\varepsilon)$ over a horizon T, because an error at step t affects all subsequent steps. For a cyclic task that spans many cycles this quadratic growth is the central obstacle. A policy that reproduces each phase well in isolation can still fail after a few cycles once small deviations in grasp height or nut angle accumulate.

DAgger [20] addresses the distribution mismatch by querying the expert on the states that the student itself visits. In iteration i the environment is driven by a mixture policy that executes the expert's action with probability β_i and the student's action otherwise. The expert's action is recorded as the label for every visited state, the labeled states are added to the aggregated dataset, and the student is retrained. Under a no-regret learner, the resulting policy has expected cost $O(T\varepsilon)$, linear rather than quadratic in the horizon. The guarantee holds because the aggregated dataset converges to the state distribution of the learned policy itself, so that the supervised error measured on that dataset is the error the policy will actually incur. Behavior cloning, in contrast, measures its error on the expert's distribution, where the policy is never evaluated. The price is the $O(T)$ expert queries per iteration and the requirement that the expert be queryable at every visited state. The schedule β_i is typically decayed from one toward zero, so that early iterations stay close to the expert's distribution and later iterations expose the student to its own mistakes. SafeDAgger [31] reduces the number of expert queries by invoking the expert only when a learned safety classifier, or a threshold on the disagreement between expert and student actions, predicts that the student is about to fail. The predecessor of this thesis applied DAgger and SafeDAgger to distill a privileged grasping teacher into an RGB student and found that the safety-gated variant produced both the best grasp success and the fewest unsafe episodes [26].

DAgger therefore requires an expert that can be queried at arbitrary student-visited states. A privileged simulation policy satisfies this requirement. A recorded demonstration does not. The hybrid teacher of this thesis is such a queryable expert, and the method of Chapter 3 uses this property in two ways: offline, by disturbing the teacher during collection so that it labels its own recoveries from off-nominal states, and online, through a DAgger variant with a flow-matching head that serves as a comparison.

2.1.4 Generative Action Heads: Diffusion and Flow Matching

A deterministic regression head predicts one action for each observation. Demonstrations of manipulation tasks are, however, often multimodal. Two demonstrators, or one teacher recovering from two different disturbances, may approach the same nut from different sides, and averaging their actions produces an action that neither of them would take. Generative action heads represent the full conditional distribution $p(\boldsymbol{u} \mid \boldsymbol{y})$ and sample from it at run time. Diffusion Policy [21] models the distribution of action sequences by iterative denoising. Gaussian noise is added to demonstrated action chunks in a forward process, a network is trained to predict the noise, and actions are generated by reversing the process over tens to hundreds of denoising steps. Diffusion policies handle multimodality and predict action chunks, which improves temporal consistency, but their sampling cost is high for control loops with tight latency budgets.

Flow matching [23], [32] reaches the same goal with a simpler training target and a cheaper sampler. Instead of a stochastic denoising process it learns a time-dependent velocity field $\boldsymbol{v}_\theta(\boldsymbol{x}, \lambda)$ that transports samples of a simple source distribution p_0, typically a standard

Gaussian, to the data distribution p_1 along the ordinary differential equation (ODE) $\dot{\boldsymbol{x}} = \boldsymbol{v}_\theta(\boldsymbol{x}, \lambda)$ for $\lambda \in [0,1]$. The conditional flow-matching objective regresses the velocity field onto the velocity of a prescribed conditional probability path between a source sample and a data sample. With the linear interpolant $\boldsymbol{x}^\lambda = (1-\lambda)\boldsymbol{x}^0 + \lambda \boldsymbol{x}^1$, whose velocity is $\boldsymbol{x}^1 - \boldsymbol{x}^0$, the objective is a plain regression, and the learned ODE can be integrated with a handful of Euler steps. Section 3.5.2 derives this objective and states it for conditional action chunks.

Diffusion and flow matching are closely related. The forward noising process of a diffusion model defines a Gaussian probability path between data and noise, and its reverse process can equally be written as a deterministic probability-flow ODE driven by the learned score. Flow matching with Gaussian conditional paths contains this diffusion path as a special case, and with the optimal-transport path it replaces the curved diffusion trajectories by straight lines that are cheaper to integrate [23]. The practical consequences are the absence of a noise schedule, a regression target that is a simple difference of two samples, and samplers with few steps. Flow matching is the action head of vision-language-action models such as π_0 [22], which generate 50-step action chunks with ten integration steps, and it is the action head of the student in this thesis. RQ2 asks whether this generative head distills the hybrid-teacher demonstrations better than deterministic regression with recurrent or Transformer backbones.

2.1.5 Visual Representations for Manipulation

A visuomotor student must extract the nut pose relative to the hand from camera data. Three families of encoders are relevant here. Image-space encoders apply a convolutional backbone such as ResNet-18 [33] to the RGB image, possibly with depth as an additional channel, and pool the feature map into a vector. They inherit ImageNet pretraining but have to learn metric geometry from appearance, and they must be made robust to texture and lighting with visual domain randomization [34]. Point-cloud encoders back-project the depth image through the calibrated camera model into a set of 3D points, optionally attach the pixel color, and encode the set with a small shared point network followed by global max pooling in the spirit of PointNet [35]. The 3D Diffusion Policy (DP3) encoder [27] is such a compact point encoder. DP3 shows that it generalizes better across viewpoints and appearance than image encoders for diffusion-based visuomotor policies, because the metric geometry enters the representation directly. Adapt3R [36] combines the two families. Dense 2D features from a frozen pretrained backbone are localized in 3D by the depth map and expressed in the end-effector frame before attention pooling, which makes the representation invariant to the embodiment and robust to domain change. All three families are implemented in the code base of this thesis. Section 3.5.3 explains why the DP3-style encoder was retained, and RQ2 compares it with the image-space encoder.

2.1.6 Operational-Space Control

Both the teacher and the student command a Cartesian end-effector target, and a task-space impedance controller converts this target to joint torques. Task-space motion control with torque inputs and impedance control are treated in standard references [37], [38], [39]. This section states the law in the form that the thesis uses. Let $\boldsymbol{q}$ be the joint positions of the seven-DoF arm, $J(\boldsymbol{q})$ the geometric Jacobian of the controlled frame, and $\boldsymbol{e} \in \mathbb{R}^6$ the position and axis-angle orientation error between target and current pose. The controller applies

$$\boldsymbol{\tau} = J^{\mathrm{T}}\big(K_p \boldsymbol{e} - K_d \dot{\boldsymbol{x}}\big) + N^{\mathrm{T}}\big(K_{p,0}(\boldsymbol{q}_0 - \boldsymbol{q}) - K_{d,0}\dot{\boldsymbol{q}}\big), \tag{2.3}$$

where $\dot{\boldsymbol{x}}$ is the end-effector twist, K_p and $K_d = 2\sqrt{K_p}$ are diagonal stiffness and critical-damping gains, and the second term pulls the redundant seventh degree of freedom toward a rest posture $\boldsymbol{q}_0$ through a nullspace projector N with posture gains $K_{p,0}$ and $K_{d,0}$. The first term is a task-space spring and damper: the end effector behaves as if it were attached to the target by a virtual stiffness K_p and damping K_d, which is the defining property of impedance control [39].

The nullspace projector decides how strongly the posture term interferes with the task. It is dynamically consistent when $N^{\mathrm{T}} = I - J^{\mathrm{T}}\bar{J}^{\mathrm{T}}$ with the inertia-weighted pseudoinverse $\bar{J} = M^{-1}J^{\mathrm{T}}(JM^{-1}J^{\mathrm{T}})^{-1}$ and joint-space inertia M [40], [41]. Torques in the nullspace then produce no acceleration of the controlled frame. A projector built from a pseudoinverse of the Jacobian without inertia weighting needs no inertia model, but it decouples the nullspace from the task only statically, so that posture torques can still disturb the task during motion [40]. The simulated controller uses the dynamically consistent projector, and the controller on the physical robot uses a damped kinematic one; Section 5.1 describes this difference. Equation (2.3) is the Factory and FORGE controller [14], [16], which this thesis inherits. Scaling the spring and damper term by the task-space inertia $\Lambda = (JM^{-1}J^{\mathrm{T}})^{-1}$ and compensating the nonlinear dynamics yields the full operational-space formulation, in which the closed loop becomes a decoupled unit-mass system in every task direction [37].

Impedance control is preferred over stiff position control for a contact-rich task because the compliant law yields on unexpected contact rather than driving through it [39], and because the policy can shape the interaction by commanding pose offsets rather than torques. The compliance is also what allows the simplified thread model of Section 3.2.2 to tolerate small deviations of the hand from the helical path.

2.2 Related Work

The related work is ordered along the three difficulties of threading. Section 2.2.1 reviews robotic assembly and threading, the application itself. Section 2.2.2 reviews learned dexterous manipulation (difficulty (b)), Section 2.2.3 reviews sim-to-real transfer for contact-rich manipulation (difficulty (a)), and Sections 2.2.4 and 2.2.5 review long-horizon tasks and the teacher–student distillation with which this thesis addresses them (difficulty (c)). Section 2.2.6 states the gap.

2.2.1 Robotic Assembly and Threading

Assembly tasks such as peg-in-hole insertion, gear meshing, connector mating, screwdriving, and nut–bolt fastening demand precise motion and closed-loop contact under geometric and dynamic uncertainty [42]. In industrial practice, threaded joints are assembled with dedicated tools such as nutrunners and screwdriving spindles, and fixtures present the parts in a known pose. Learning-based assembly in simulation aims to relax these requirements. Factory [14] made nut-and-bolt assembly learnable by combining signed-distance-field collision detection, contact reduction, and a Gauss–Seidel solver, so that a thousand parallel assemblies run in real time on one GPU. It also splits the task into pick, place, and screw subpolicies. FORGE [16] adds force-guided exploration under pose uncertainty and demonstrates a multistage planetary-gear assembly that includes nut threading, and IndustReal [17] transfers simulation-trained assembly policies to a physical arm. All three use a parallel-jaw gripper.

Screwdriving with hand tools has been modeled analytically and learned with physics-informed models [8], [9], [43]. With multi-fingered hands, unscrewing a bottle lid, a

threaded joint released by turning, has been learned with two hands by sim-to-real RL [44] and with one hand from teleoperated demonstrations with a visuomotor diffusion policy [45]. Play2Perfect [19] pretrains a goal-conditioned manipulation policy through play and fine-tunes it for precise assembly, including screwing a rod-like furniture leg. This gives goal-directed screwing in a structured setting, but not for everyday threaded parts such as nuts. The closest prior work is DexScrew [18], which uses a hand–arm setup comparable to the one here and replaces the thread geometry by a revolute-joint proxy to learn a rotational finger gait. The gait assists teleoperation and supports the collection of real demonstrations for tactile behavior cloning. Its policy spins the fastener without a target angle and without learned termination. Goal-directed threading over many grasp-and-release cycles, which this thesis addresses for several nut sizes, is left open by these works.

2.2.2 Dexterous Manipulation with Learned Policies

A multi-fingered hand removes the need for task-specific tooling, but it brings high-dimensional control and complex contact dynamics with it [4], [5]. Analytical models of motion and contact have rarely coped with the uncertainty of real environments, and the field has therefore turned to learning [5]. Model-free RL in simulation learns in-hand reorientation of more than two thousand objects [6]. DeXtreme [7] transfers agile in-hand reorientation to a low-cost hand with a vision-based pose estimator, and DexPBT [13] learns regrasping and grasp-and-throw for hand–arm systems with population-based training. DextrAH-G [25] and DextrAH-RGB [24] grasp a wide range of objects with a hand–arm system from depth and RGB images, and SimToolReal [10] manipulates tool-like objects to random goal poses and generalizes zero-shot to real tools. Without learning, finger gaits that pivot an object under contact-force control have been demonstrated [46].

These works name the high-dimensional action space of the hand as a central challenge [15], [25]. Most of them solve one skill per episode: a grasp, a goal pose of a tool, or a target orientation. In-hand reorientation chains goals and counts consecutive successes [7], but the object stays in the hand. Threading additionally requires the hand to release the part and acquire it again while the bolt constrains it, and the grasp must hold under the turning load in between. This thesis narrows the solution space to one grasp strategy, a thumb–index pinch, and lets the learned policy control this pinch together with the arm (Section 3.2).

2.2.3 Sim-to-Real Transfer for Contact-Rich Manipulation

Policies trained in simulation meet discrepancies in perception, dynamics, contact, and control when they are deployed on hardware, and the discrepancies grow with the dimensionality of the control problem and the amount of contact involved. Thread contact is an extreme case, because engaged threads produce thousands of simultaneous contacts [14]. The most common response is to make the policy robust enough for zero-shot transfer, by randomizing visual and physical parameters during training [7], [34], by randomizing dynamics together with a force threshold [16], or by making the visual representation itself transferable [36]. A second family adapts a simulation-trained policy with real-world data, for instance by tuning the simulator to a few minutes of real data [47], by learning a residual policy from on-robot corrections [48], or by continuing training on the robot [49], [50].

This thesis follows the first family, and RQ3 asks whether it suffices for threading. The student is trained on simulated data only, and transfer rests on four properties of the training setup. The camera model is calibrated against the physical cell and reproduced in simulation. The visual representation is a metric point cloud in the spirit of DP3 [27], which is

invariant to appearance up to depth noise. The rendered appearance is randomized in the style of DextrAH-RGB [24]. The Cartesian action is executed on the robot by a controller that reproduces the simulated control law. The approach requires no real-world data collection and no training on the robot.

2.2.4 Long-Horizon Tasks and Policy Chaining

A single policy for a long-horizon task faces two difficulties. Small control errors compound over the horizon [20], and fine-tuning the policy on later subtasks can erase what it learned on earlier ones, the catastrophic forgetting problem [51]. Larger networks, frozen or regularized parameters, and architectures with longer temporal memory such as recurrent networks and Transformers mitigate these effects but do not remove them. At large scale, a single policy can nevertheless be trained end to end. AlphaNIST [52] trains policies from scratch with a sparse binary task reward and without demonstrations, progress shaping, or scripted primitives, and its policies complete ten assembly pairs in one simulated episode without a prescribed order.

The alternative is to decompose the task into skills with their own policies and to chain them. Each skill is then easier to learn, but the chain introduces a transition problem. The states in which one skill terminates may lie outside the initial-state distribution on which the next skill was trained, so the chain breaks at the boundary. Adversarial skill chaining [53] regularizes the terminal-state distribution of each skill toward the initial-state distribution of its successor. Sequential Dexterity [15] learns a transition-feasibility function with which adjacent dexterous policies are fine-tuned, switches between them autonomously, and recovers from failed transitions. A cyclic task sharpens the transition problem, because the same boundaries recur in every cycle while the task state advances, so every transition error has many chances to occur and its effect carries into the next cycle.

The method of this thesis decomposes only during data generation, where every phase has a simple specification, and unifies at deployment by distilling the chained demonstrations into one policy. The transitions are then part of what the student learns rather than a problem to be solved between separate policies. RQ1 tests whether this unified student sustains consecutive cycles better than a single-cycle policy that is invoked repeatedly.

2.2.5 Teacher–Student and Visuomotor Distillation

Learning by Cheating [54] established the route from a policy that is trained with privileged simulator state to a deployable policy that acts from onboard sensing. In dexterous manipulation, DextrAH-G [25] and DextrAH-RGB [24] distill a privileged RL grasping teacher into depth- and RGB-based students with online DAgger. UniGraspTransformer [55] consolidates many object-specific teachers into one student, and Lin et al. [47] learn separate single-task RL policies on a humanoid and distill them into one generalist vision-based policy. The author's semester thesis [26] applied this recipe to a 23-DoF hand–arm system and compared behavior cloning, DAgger, and SafeDAgger.

For long-horizon behavior, two further ingredients matter. Predicting a sequence of future actions instead of a single action improves temporal consistency. Action Chunking with Transformers (ACT) [56] predicts chunks with a Transformer and blends overlapping chunks by temporal ensembling. Representing the action distribution generatively lets a policy keep several valid behaviors apart instead of averaging them. Diffusion Policy [21] models multimodal action sequences by iterative denoising, DP3 [27] conditions it on compact point-cloud features, and flow matching [23] learns a continuous vector field that generates actions by

integrating an ODE in few steps, as the vision-language-action model π_0 [22] does. The student of this thesis combines a point-cloud encoder, action chunking with temporal ensembling, and a flow-matching head, because its training data mix nominal teacher actions with the teacher's recoveries from disturbances.

2.2.6 Summary and Research Gap

The related work addresses each difficulty of threading separately. Simulation-based assembly learning makes thread contact tractable and transfers to hardware, but with a parallel-jaw gripper and a fastener that is already held [14], [16], [17]. Learned dexterous threading and screwing covers a spinning gait for an engaged fastener [18] and goal-conditioned screwing of a furniture leg [19]. Policy chaining reaches long horizons at the cost of transitions between separately deployed policies [15], [53], and teacher–student distillation yields deployable visuomotor policies for single skills [24], [25]. No prior work threads a nut over many cycles with learned grasping and re-grasping on a dexterous hand. Chapter 3 closes this gap by combining the concepts of Section 2.1: a simplified thread model makes RL in simulation feasible, a privileged teacher learns the pinch-grasp turn, scripted transitions chain it into multi-cycle demonstrations, and one flow-matching student distills these demonstrations for zero-shot deployment.

Chapter 3

Methodology

This chapter presents the method. Its parts follow the three difficulties of Section 1.2, and each part answers one of them. Section 3.1 formalizes cyclic nut threading and the metrics that evaluate it. Section 3.2 describes the simulated and physical system: the hand–arm platform, the simplified thread model that makes learning in simulation tractable (difficulty (a)), and the action contract with its controller. Section 3.3 presents the privileged reinforcement-learning teacher that learns to grasp and turn the nut with the dexterous hand (difficulty (b)). Section 3.4 explains how the hybrid teacher chains this skill with scripted transitions into complete multi-cycle demonstrations, which is the data side of difficulty (c), and how the pipeline prepares the student for zero-shot transfer, which is the transfer side of difficulty (a). Section 3.5 develops the central contribution, the distillation of these demonstrations into a single visuomotor student by conditional flow matching, which answers difficulty (c) at deployment, and relates the offline and online variants. Section 3.6 defines the evaluation protocol with which Chapters 4 and 5 answer RQ1 to RQ3. Figure 3.1 gives an overview.

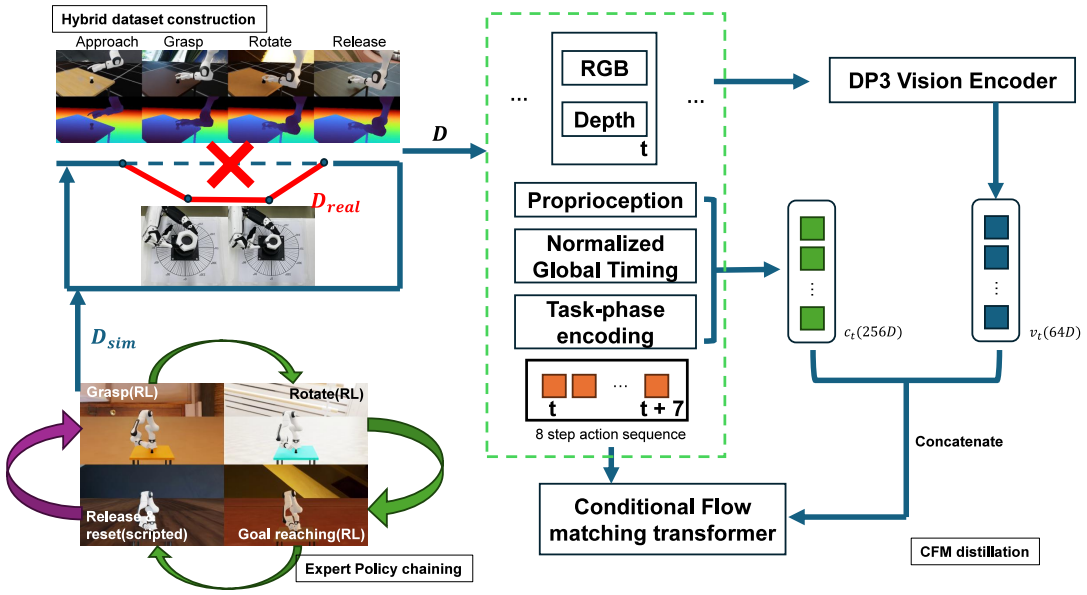


Figure 3.1: The pipeline in detail. Left: the privileged RL policy executes grasp, rotation, and goal reaching and a scripted controller executes release and reset; chained in a loop, they form the hybrid teacher that generates the simulated demonstrations $\mathcal{D}_{\text{sim}}$. Right: the student encodes the aligned RGB-D observation into a 64-D point-cloud feature, fuses it with proprioception, the normalized global progress, and the process-phase encoding into a 256-D condition, and a conditional flow-matching Transformer predicts an eight-step action chunk.

3.1 Problem Formulation

Cyclic nut threading is decomposed into the ordered phase sequence

$$\mathcal{S} = \big(\underbrace{\text{approach, grasp, rotate, goal reaching}}_{\mathcal{M}_{\text{RL}}:\,\pi_{\text{T}}},\ \underbrace{\text{release, return and reset}}_{\mathcal{M}_{\text{script}}:\,\kappa}\big). \tag{3.1}$$

The active phase at control step t is $m_t \in \mathcal{M} = \mathcal{M}_{\text{RL}} \cup \mathcal{M}_{\text{script}}$. The four phases in $\mathcal{M}_{\text{RL}}$ are executed by a learned teacher policy π_{T}; the two transition phases in $\mathcal{M}_{\text{script}}$ are executed by scripted controllers κ_m. The split is a property of the data-generation stage only: the deployed student receives neither the phase label as a switch nor separate networks per phase.

Let K be the required number of cycles, $\Delta\varphi^*$ the desired clockwise rotation per cycle, and φ_k the unwrapped nut angle about the bolt axis after the goal-reaching phase of cycle k, with φ_0 the angle at the start of the trial. The cyclic tracking error, the episodic tracking error, and the tracking cost of a trajectory τ are defined as

$$e_k = \varphi_k - \varphi_{k-1} - \Delta\varphi^*, \qquad e_{\text{ep}} = \varphi_K - \varphi_0 - K\Delta\varphi^* = \sum_{k=1}^{K} e_k, \qquad E(\tau) = \frac{1}{K}\sum_{k=1}^{K} e_k^2 + \beta_E\, e_{\text{ep}}^2. \tag{3.2}$$

Positive errors indicate overshoot and negative errors lag; the cost squares them so that overshoot in one cycle cannot cancel lag in another, and the weight β_E trades accumulated drift against cycle-wise accuracy. Let $C(\tau) \in \{0, \dots, K\}$ count the cycles completed before failure or timeout and let $\Pi_{\mathcal{S}}$ denote the policies that execute the phases of $\mathcal{S}$ in order in every cycle. The research problem is to find the policy that chains all phases with maximum completion success and, among those, minimum tracking cost:

$$\Pi_{\text{succ}} = \arg\max_{\pi \in \Pi_{\mathcal{S}}} \Pr_{\tau \sim p_\pi}[C(\tau) = K], \qquad \pi^* = \arg\min_{\pi \in \Pi_{\text{succ}}} \mathbb{E}_{\tau \sim p_\pi}[E(\tau) \mid C(\tau) = K]. \tag{3.3}$$

Throughout this thesis $K = 10$ and $\Delta\varphi^* = 60°$, which corresponds to one face of a hexagonal nut per cycle. The policy π^* is learned from fixed hybrid-teacher demonstrations rather than by optimizing (3.3) on hardware, because on-robot optimization of a contact-rich cyclic task is neither safe nor sample-efficient.

The task is partially observable at deployment. The simulator state $\boldsymbol{x}_t$ contains the nut pose, the contact state, and the controller parameters; the student observes only an aligned RGB-D image, its proprioception, and two low-dimensional conditioning signals introduced in Section 3.5.3. The teacher–student structure of the method separates the two difficulties of this POMDP: the teacher solves the control problem with privileged state, and the student learns to perceive and to reproduce the teacher's behavior from sensor data.

3.2 Simulated and Physical System

The method is trained in simulation on a model of the physical hand–arm system and deployed on that system. This section describes the platform, the thread model that makes learning in simulation tractable, and the action contract that teacher, student, and robot share.

3.2.1 Platform

The robot is a seven-DoF torque-controlled arm with a five-fingered right hand mounted on the flange through a 10 mm adapter; Section 5.1 names the products. The hand has twelve

finger joints driven by six motors. The non-thumb fingers each have one actuated flexion joint with a mechanically coupled distal joint, and the thumb has an abduction joint and a coupled flexion chain. Only three hand degrees of freedom are commanded by the policy: thumb abduction, thumb flexion, and index flexion, which together form the pinch used for threading. The remaining fingers hold a curled posture. This restriction narrows the solution space of the dexterous hand to one grasp strategy, as difficulty (b) of Section 1.2 requires. The nut is threaded onto a bolt that stands on a 5 cm pedestal on a 60 cm by 80 cm bench. The robot base is mounted at tabletop height, 46.5 cm from the near table edge, so that the reach envelope lies above the work surface. A fixed RGB-D camera provides the observation; its extrinsic calibration relative to the robot base was measured on the physical cell and is reproduced in simulation.

Simulation uses Isaac Sim 5.1 with Isaac Lab 2.3 on the PhysX backend. Physics runs at 120 Hz; both teacher and student act at 15 Hz, so one policy command is held over eight physics substeps while the low-level controller closes its loop at the physics rate. The platform was selected over an earlier humanoid-arm platform on which the project started, because the present arm is substantially more precise and repeatable on the physical system. The task code, rewards, observation contracts, and training recipe are identical for both robots; only the robot asset, workcell geometry, and reset poses differ.

3.2.2 Simplified Thread Simulation

The thread model is the first answer to difficulty (a) of Section 1.2: the skill is learned in simulation, so the simulated thread has to be cheap enough for the thousands of parallel environments that PPO needs. Simulating a mating thread with mesh colliders is prohibitively expensive at that scale. A kinematic thread model also gives the policy a well-posed learning signal: with signed distance field (SDF) thread colliders, axial progress of the nut can be obtained by pressing it along the bolt axis, whereas a kinematic coupling makes axial progress a consequence of rotation alone, so that progress can only be earned by turning. The thread model used for teacher training therefore replaces the thread contact by a kinematic coupling, illustrated in Figure 3.2. The nut is attached to the bolt by a passive revolute joint about the bolt axis and a coaxial, position-controlled prismatic joint. With prismatic coordinate d_t, unwrapped and direction-corrected revolute coordinate θ_t, thread pitch ρ, and the index b_k of the step at which the current turn started, the prismatic target is

$$d_t^* - d_{b_k} = -\frac{\rho}{2\pi}\left(\theta_t - \theta_{b_k}\right), \tag{3.4}$$

so that one revolution produces one pitch of axial travel. The prismatic joint tracks d_t^* with a stiff drive (stiffness 2×10^5 N/m, damping 200 N s/m, force limit 1000 N), whose finite stiffness permits small deviations from the ideal helix while the revolute–prismatic pair constrains lateral motion and keeps the nut near the commanded path. The finger–nut contact forces remain fully simulated: the policy must physically grip and turn the nut. Thread resistance is modeled on the revolute joint by a smoothed Coulomb and viscous friction torque

$$\bar{\tau}_{f,t} = -\tau_c \tanh\left(\frac{\omega_t}{\omega_s}\right) - c_v\,\omega_t, \tag{3.5}$$

with revolute velocity ω_t, Coulomb torque $\tau_c = 2 \times 10^{-3}$ N m, viscous coefficient $c_v = 2 \times 10^{-4}$ N m s/rad, and smoothing speed $\omega_s = 0.1$ rad/s. The nut carrier has a mass of 1 g, and the joint limits reserve one full turn above the reset pose and a margin above the physical lower stop. The pitches are 3.0 mm, 3.5 mm, and 4.0 mm for the M24, M30, and M36 nuts.

The model has one deliberate simplification beyond contact: the nut is initialized already engaged with the bolt. Thread engagement from a loose nut is a separate skill with a narrow entry configuration, and it is outside the scope of this thesis. The simplified contact opens a gap between the simulated and the physical thread, which the transfer design of Section 3.4.4 has to bridge.

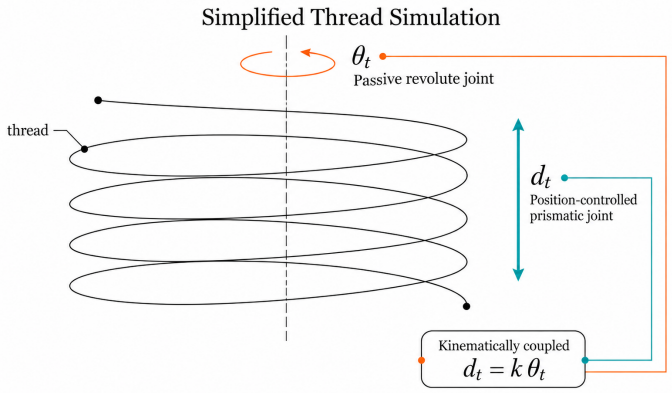


Figure 3.2: Simplified thread simulation. A passive revolute joint θ_t about the bolt axis is kinematically coupled to a position-controlled prismatic joint d_t through $d_t = k\theta_t$ with $k = -\rho/(2\pi)$, so that the nut advances by one pitch per revolution without explicit thread contacts; the finger–nut contacts remain fully simulated.

3.2.3 Action Contract and Controller

Teacher and student share one action space, a 9-D vector $\boldsymbol{a}_t \in [-1,1]^9$ that is decoded into a Cartesian target for the hand's grasp frame and three pinch-joint targets. The grasp frame is a fixed offset from the flange whose origin lies at the thumb–index pad midpoint, whose z axis points along the approach direction, and whose x axis is the pinch-closing axis. The first three components scale by 0.05 m to a position offset relative to an action frame that is the bolt tip during threading and the held nut during pickup; because the target is expressed relative to the bolt, the controller draws the grasped nut toward the bolt regardless of where nut and robot are, and a zero action does not hold the current pose. Components four to six set the orientation: roll and pitch scale by 1 rad around a neutral grasp orientation, and yaw maps $[-1,1]$ to the window $[-180°, 90°]$ inside the wrist joint limits. Components seven to nine command the thumb abduction, thumb flexion, and index flexion joints, where zero holds the closed-grasp posture and ± 1 moves by half the admissible joint range. The raw action is smoothed by an exponential moving average with factor sampled per episode from $[0.025, 0.1]$ (three times larger, capped at one, for the fingers), and the resulting target is clipped to at most 0.02 m and 0.097 rad from the current pose per step, so that one command cannot demand a large jump.

The Cartesian target is tracked by the task-space impedance controller of (2.3), inherited from Factory and FORGE, with proportional gains $[565, 565, 565, 28, 28, 28]$ (N/m and N m/rad), critical damping, a nullspace posture term with gains 10 and 6.32, and a dead zone of $[5, 5, 5, 1, 1, 1]$ (N and N m) below which wrench components are zeroed. The gains act component-wise in world axes, so the impedance ellipsoid is aligned with the vertical bolt axis. The arm's own joint drives are disabled and the controller output is applied as joint torque; the finger joints are position-controlled by the simulator's implicit PD drives. For training robustness and for sim-to-real transfer, the controller parameters are randomized per episode: gains by a multiplicative factor in $[1/1.41, 1.41]$, position and rotation thresh-

olds by ±25% and ±29%, and the dead zone by uniform resampling every two seconds. The critic observes the sampled values, the actor does not.

For deployment, the same controller law is reproduced on the physical arm in a Cartesian impedance controller that uses the quaternion-vector orientation error, a damped pseudoinverse for the nullspace projector, and the robot's own gravity and Coriolis compensation; the gains of the physical controller are documented in Chapter 5. The shared Cartesian action space for teacher, student, and robot results from a design decision, and the decision is recorded here because the project also implemented and evaluated an alternative. The alternative, called the 17-D impedance joint-PD contract, has the student emit seven arm position references, seven arm velocity references, and three pinch positions, which a joint-space impedance controller tracks as

$$\tau = g(q) + K_p^{\mathrm{hw}}(q_{\mathrm{ref}} - q) + K_d^{\mathrm{hw}}(\dot{q}_{\mathrm{ref}} - \dot{q}), \tag{3.6}$$

with gravity compensation $g(q)$ owned by the controller. Teacher labels are converted to this contract by evaluating the Cartesian controller and reverse-engineering the equivalent joint references, $q_{\mathrm{ref}} = q + (K_p^{\mathrm{hw}})^{-1}\tau_{\mathrm{OSC}}$ and $\dot{q}_{\mathrm{ref}} = \dot{q}$. The motivation was that a joint-space command is the native interface of most robot drivers. Replaying recorded 17-D commands at the physics rate reproduced the teacher motion with an arm-joint RMSE of 0.0037 rad, so the controller itself is adequate. The contract is nevertheless unfavorable for learning, for three reasons. First, the equivalent joint target changes within the eight physics ticks over which a 15 Hz student command is held, so a zero-order hold introduces a target error of 0.010 to 0.015 rad and an implied torque error of up to 16 N m. Second, the commands are in physical units with widely different scales across the 17 dimensions. Third, as Chapter 4 reports, only students trained on the Cartesian contract reach the task goals in closed loop. The Cartesian contract, in which the student imitates the bounded, bolt-relative action that the teacher itself produced and executes it through the unchanged controller, removes this conversion and is the contract of all reported results. The 17-D contract is retained as an ablation.

3.3 Privileged Teacher Training

The privileged teacher answers difficulty (b) of Section 1.2. It learns to approach the nut, acquire the pinch grasp, keep it under load while turning, and hold the target, all through the commanded hand and arm degrees of freedom of Section 3.2.1. Because it observes privileged simulator state, it solves this control problem without having to perceive the nut (Section 3.1).

The teacher for the learned phases in $\mathcal{M}_{\mathrm{RL}}$ is a recurrent policy trained with PPO in an asymmetric actor–critic. The actor observation $z_t^{\pi} = g_{\pi}(x_t, \xi_t)$ selects the actor-visible components of the simulator state and applies observation noise ξ_t; it comprises the fingertip (grasp-frame) position relative to the bolt frame and its quaternion, the end-effector linear and angular velocity obtained by finite differences, the fingertip position relative to the held nut and the nut quaternion, the position and quaternion error to the current turning target, a grasp-phase flag that indicates whether the proper grasp has latched, the previous filtered action, and, for the multi-size teacher, a one-hot code of the nut size. Position terms receive Gaussian noise with 0.25 mm standard deviation per axis, orientation terms 0.1°, and the bolt frame a per-episode offset of 1 mm, which models calibration error. The critic observes the same task and history structure using clean values and extends it with the arm joint positions, the complete nut and bolt pose, the sampled controller gains, smoothing factor,

and thresholds, and the thumb and index contact forces. Table A.1 in the appendix lists the exact components and widths.

Actor and central critic share the same architecture: a two-layer LSTM with 1024 units per layer and layer normalization, followed by a 512–128–64 MLP with ELU activations. Recurrent sequences span 128 control steps, the rollout horizon is 256 steps, and the actor parameterizes a diagonal Gaussian over the normalized action with a state-dependent standard deviation. The teacher is optimized with the shaped reward

$$r_t = \sum_{i \in \mathcal{R}} w_i^r R_{i,t} - \sum_{j \in \mathcal{P}} w_j^p P_{j,t},$$
$$\mathcal{R} = \{\text{center}, \text{face}, \text{grasp}, \text{turn}, \text{target}\}, \qquad \mathcal{P} = \{\text{mag}, \Delta a, \text{align}, \text{force}, \text{settle}, \text{clear}, \text{contact}\}. \tag{3.7}$$

The reward follows the three phases of the learned behavior. During approach, R^{center} rewards motion of the grasp center toward the nut center and R^{face} rewards motion of the thumb and index pads toward the two designated opposing faces of the nut; both pay 20 times the newly achieved fraction of the initial gap, so repeated motion cannot accumulate reward. R^{grasp} pays a one-time bonus of 100 when a proper grasp, defined as simultaneous thumb and index contact on the designated faces with a pinch line perpendicular to the nut axis, has held for five consecutive steps. During turning, R^{turn} pays 60 times the newly achieved fraction of the distance to the fixed helical target pose, and pays only while the previously acquired proper grasp remains live. At completion, R^{target} pays 250 distributed over a five-step hold within 5° of the target. The penalties discourage large commands (P^{mag}, weight 10^{-3}), abrupt action changes ($P^{\Delta a}$, 0.1), misalignment of the hand frame with the nut frame (P^{align}, 0.02), excess wrist force (P^{force}, 0.05), residual rotation near the target (P^{settle}, $0.25\,\omega_t^2$ from 50° onward), insufficient table clearance (P^{clear}), and arm–table contact (P^{contact}). An episode succeeds when at least 55° of clockwise rotation is held with a live proper grasp for five steps; it terminates early when the nut leaves the bolt by more than 3 cm, tilts by more than 60°, or releases the thread, and when the grasp is lost after it has latched. The last rule turns grasp retention into a hard constraint rather than a shaped term.

Robustness is obtained by domain randomization of the nut mass (±5 g), the bolt friction (static coefficient in $[0.25, 1.25]$), the controller parameters of Section 3.2.3, and the initial state; Table 3.1 lists the ranges. The observation noise is small in absolute terms, a quarter of a millimeter and a tenth of a degree, because the actor's inputs are relative poses that the student will later infer from a calibrated camera; the per-episode bolt-frame offset of one millimeter is the term that models calibration error and forces the teacher to close the loop on the nut rather than on the bolt frame. The initial nut rotation is randomized by an automatic curriculum: the yaw range starts at zero and expands by 5° whenever the population success rate exceeds 80 %, up to ±30°, with at least five episode lengths between expansions; the initial axial height is sampled uniformly up to 20 mm below the nominal spawn while reserving room for the full 60° target turn.

A single teacher is conditioned on the M24, M30, and M36 nuts through the one-hot code and round-robin assignment of sizes to environments. Training uses 512 parallel environments for 200 epochs and takes about ten hours on one NVIDIA GeForce RTX 5090; the PPO hyperparameters are listed in Table A.2 in the appendix.

A second expert, the pickup teacher, was trained with the same recipe for grasping a loose nut from the table, lifting it, transporting it to a handoff pose above the bolt tip, and engaging it; its reward uses the same approach and grasp terms, a lift term, and keypoint progress toward two waypoints. It is part of the full-task variant of the hybrid teacher described below and is reported for completeness in Chapter 4; it is not part of the final threading pipeline, whose trials start from an engaged nut.

Table 3.1: Domain randomization during teacher training. $U(a, b)$ denotes uniform sampling; multiplicative factors are sampled as $1 + U(0, \ell)$ or its reciprocal with equal probability.

Group	Quantity	Range
physics	nut mass offset	$U(-5, 5)$ g per reset
	bolt static friction	$U(0.25, 1.25)$, 128 buckets
controller	proportional gains	multiplicative, $\ell = 0.41$, per axis and reset
	action smoothing factor	$U(0.025, 0.1)$ per reset; fingers $3\times$
	position / rotation threshold	multiplicative, $\ell = 0.25$ / 0.29
	dead zone	$U(0, [5, 5, 5, 1, 1, 1])$ N, N m, resampled every 2 s
observation	fingertip and nut position	$\mathcal{N}(0, 0.25\,\text{mm})$ per axis and refresh
	fingertip and nut orientation	random-axis angle $\mathcal{N}(0, 0.1^\circ)$
	bolt frame	$\mathcal{N}(0, 1\,\text{mm})$ per axis and reset
	fingertip forces	$\mathcal{N}(0, 1\,\text{N})$ per channel
initial state	nut yaw	curriculum $0 \rightarrow \pm 30^\circ$ in 5° steps at 80 % success
	nut axial height	$U(0, 20)$ mm below nominal, full turn reserved
	nut size	round-robin M24 / M30 / M36 (multi-size teacher)

3.4 Hybrid Teacher and Demonstration Generation

The privileged teacher turns the nut once. The hybrid teacher answers the data side of difficulty (c) of Section 1.2: it chains the learned turn with scripted release and return, so that each phase is specified by the mechanism that specifies it most simply, and it rolls the chain out in simulation to produce complete multi-cycle demonstrations. This section describes the cyclic rollout, its full-task variant, the disturbance augmentation, and the dataset. It also collects the design choices that answer the transfer side of difficulty (a) (Section 3.4.4), because most of them are fixed when the demonstrations are rendered.

3.4.1 Cyclic Teacher Rollout in Simulation

Let $m_t \in \mathcal{M}$ denote the active phase, $\boldsymbol{x}_t$ the full simulator state, $\boldsymbol{z}_t^\pi$ the noisy actor observation, and $\boldsymbol{h}_t$ the actor's recurrent state. During rollout the teacher is queried for its deterministic mean action $\bar{\pi}_\mathrm{T}(\boldsymbol{z}_t^\pi, \boldsymbol{h}_t) = \mathbb{E}_{\boldsymbol{a}\sim\pi_\mathrm{T}(\cdot|\boldsymbol{z}_t^\pi, \boldsymbol{h}_t)}[\boldsymbol{a}]$. The hybrid teacher is

$$\pi_\mathrm{HT}(\boldsymbol{x}_t, m_t) = \begin{cases} \bar{\pi}_\mathrm{T}(\boldsymbol{z}_t^\pi, \boldsymbol{h}_t), & m_t \in \mathcal{M}_\mathrm{RL}, \\ \kappa_{m_t}(\boldsymbol{x}_t), & m_t \in \mathcal{M}_\mathrm{script}, \end{cases} \tag{3.8}$$

where κ_{m_t} is the scripted controller of the active transition phase. Both branches produce commands in the same filtered, bolt-relative Cartesian action space, so the recorded label is a single 9-D signal throughout the cycle; the scripted Cartesian targets are inverse-decoded through the action mapping of Section 3.2.3, and the executing adapter compensates the action smoothing analytically so that the controller receives exactly the requested target. This unified label space is what makes the demonstration a coherent supervision signal for one student: at no step does the label change meaning or units.

Release is triggered when the nut rotation lies within $\pm 5^\circ$ of the 60° target while a valid opposed-finger grasp has been maintained for a short dwell. The fingers then open to a safe clearance, after which four Cartesian waypoints retract the hand along the ray from the nut center through the grasp center, raise it, and return it to the initial grasp pose. Each waypoint segment is interpolated over 0.7 s and the final return over 1.0 s; the return is physically gated and counts as complete only when the grasp frame is within 5 mm and 5° of the reset frame

and the hand joints within 0.15 rad of their reset values, with a 16 s timeout that marks the cycle as a transition failure. The waypoints were authored once by a collision-safe retargeting from a captured release pose and verified to produce zero pad–nut contact force during every release. While the hand is away from the nut the thread joint is held, so the nut cannot drift between cycles. On a completed return the turning reference is rebased: the target angle advances by 60° relative to the current nut pose, the frozen threading policy receives a cycle-local canonicalized nut orientation so that it sees the same observation statistics in every cycle, and its recurrent state and action history are reseeded. The physical nut pose is not reset, so the demonstration contains the axial advance of the nut over ten cycles. The teacher population was first qualified for one period only; repeated cycles are a property of the hybrid rollout, not of the RL teacher alone. Figure 3.3 shows the coordinator that sequences one cycle.

Two details of the coordinator matter for the quality of the labels. First, the scripted transitions never write physics state: a transition may request that the thread joint be held while the hand is away, but it repairs nothing by teleporting the nut, so every recorded frame is a physically consistent consequence of the recorded commands; an earlier implementation that snapped the nut back at the policy rate produced visible reversals of the nut at release that the student would have imitated. Second, the release gate is physical rather than timed: release is decided from the measured nut rotation and the measured grasp, and the return is validated against the reset frame, so the phase labels in the recording coincide with physical events. Both properties are what allow the recorded phase to be used as a conditioning signal at deployment, where it is derived from the same physical events.

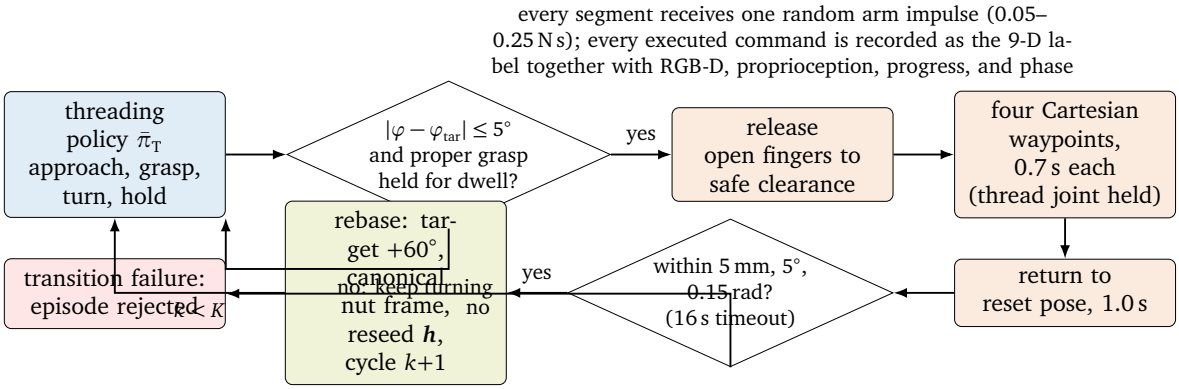


Figure 3.3: Data-collection loop of the hybrid teacher for one threading cycle. The learned policy owns approach, grasp, turn, and hold; a physical release gate hands over to the scripted release, four retreat waypoints, and the gated return; a completed return rebases the turning target and starts the next cycle. Every executed command is recorded in the common 9-D label space.

3.4.2 Full-Task Variant with Pickup and Handover

The hybrid teacher also has a full-task variant that prepends the pickup expert and a scripted pickup-to-threading handover (open the hand, move to the threading reset pose, close, settle, with phase durations of 0.4, 1.5, 0.4, and 0.25 s) to the cyclic threading teacher. The resulting demonstrations start from a loose nut on the table. Both variants and their single-phase special cases (pickup only, one threading turn, cyclic threading) are selected by one training-environment option and expose the same supervision contract, so that a student trained on either variant can be evaluated with the same tools. Because thread engagement is outside the scope of this thesis (Section 3.2.2), the reported threading results use the cyclic variant

from an engaged nut.

3.4.3 Disturbance Augmentation

Twenty demonstrations contain too little variation for a student that must recover from its own deviations. Trajectory diversity is therefore increased at collection time by applying random impulses to the arm: in every contiguous teacher-phase segment, that is, in every policy, waypoint-following, and return segment of every cycle, one impulse with a magnitude of 0.05 to 0.25 N s is applied to a random arm link between 0.05 and 0.20 s after the segment begins, giving 30 impulses per ten-cycle demonstration. The teacher, which is a closed-loop controller, reacts to the disturbance, and the recorded recovery is part of the label. Because the teacher runs deterministically at $\beta = 1$ with the disturbance as the only source of variation, the recovery segments remain consistent with the trajectory that conditions them. Demonstrations in which the teacher fails to complete all cycles are rejected before materialization; the retained set consists only of explicitly successful episodes.

3.4.4 Zero-Shot Sim-to-Real Transfer

The student is trained on the simulated demonstrations alone, $\mathcal{D} = \mathcal{D}_{\text{sim}}$, and deployed on the physical robot without real-world training data. Section 3.2.2 answered the simulation side of difficulty (a); this subsection answers its transfer side. The simplified thread model and the rendered images open a gap in contact and perception between the two domains, and four properties of the pipeline are designed to carry the student across it. RQ3 asks whether they suffice, and Chapter 5 tests it.

The first is a shared camera model. The extrinsic pose of the physical camera relative to the robot base is measured on the physical cell by an eye-to-hand calibration, and the calibrated pose and intrinsics are used to render the simulated camera; the 640×480 stream is center-cropped to 640×360 and downscaled by two in both domains, with the intrinsics scaled accordingly. Simulated and real point clouds are therefore expressed in the same frame and have the same resolution and field of view.

The second is a metric representation. The student encodes a point cloud back-projected from the depth image (Section 3.5.3), not the image itself. Points are kept only inside a fixed crop box around the workspace and within a valid depth range, so that the background, which differs most between the rendered and the physical scene, never enters the representation, and the geometry of nut, bolt, and hand is identical in both domains up to depth noise and calibration error. Random sampling of 4096 points per frame during training acts as an additional augmentation of the point set.

The third is visual domain randomization at collection time. The demonstrations are rendered with randomized dome lighting, tabletop texture, nut and bolt textures, floor texture, and image background in the style of DextrAH-RGB [24], each component switchable individually. Because the offline loader cannot regenerate scene appearance from stored images, this variation is recorded into the dataset once and reused unchanged by every student trained on it. During training, the stored RGB images are further perturbed by saturation, contrast, and brightness factors in $[0.5, 1.5]$, hue shifts in $[-0.15, 0.15]$, and motion blur with probability 0.1. The depth channel is not modified by the offline trainer; the encoder pretraining of Section 3.5.3 provides a depth-corruption model of a commodity RGB-D sensor with Gaussian and range-dependent axial noise, low-frequency spatial noise, millimeter quantization, dropout at depth discontinuities, correlated missing regions, and uniform dropout.

The fourth is a shared action path. The student imitates the teacher's bounded, bolt-relative Cartesian action, and on the physical robot this action is executed by a Cartesian impedance controller that reproduces the simulated control law (Section 3.2.3). The teacher itself was trained under randomized controller gains, smoothing factors, and dead zones and under noise on every pose input (Table 3.1), so the behavior that the demonstrations record is robust to the controller and calibration deviations that the physical system introduces.

3.4.5 Dataset Format

Each demonstration is materialized as one episode file with aligned 15 Hz samples. A sample holds the RGB image and the metric depth image at 320×180 pixels, obtained by center-cropping the 640×480 camera stream to 640×360 and downscaling by two with the intrinsics scaled accordingly; the 29-D proprioception, consisting of the ten controlled joint positions, their velocities, and the previous filtered 9-D action; the executed 9-D command; the normalized task progress; the active process phase; the nut pose for privileged ablations and auxiliary supervision; and the timestamp. The recorder stores the 17-D joint-PD equivalent of every command alongside, so that both action contracts can be trained from one collection. Chunks that cross a physical reset are masked, phase boundaries are retained so that the student learns the transitions, and terminal chunks are padded with the final valid command so that the last observations of an episode remain conditioning states. For the retained M24 dataset, 20 successful ten-cycle demonstrations collected from 32 parallel environments contain 19 486 frames.

3.5 Student Policy Distillation by Conditional Flow Matching

The student answers the deployment side of difficulty (c) of Section 1.2. The hybrid teacher cannot be deployed, because both its learned policy and its scripted controllers act on privileged simulator state (3.8). The student replaces it with one visuomotor policy that executes every phase of every cycle from sensor data, without a phase switch or separate networks. RQ1 asks whether this unified student sustains consecutive cycles better than a single-cycle policy invoked repeatedly. RQ2 asks which student design distills the demonstrations best; the alternative encoders of Section 3.5.3 and the regression baselines of Section 3.5.6 are its comparisons.

3.5.1 From Behavior Cloning to Online Distillation for Cyclic Tasks

Student learning is posed as supervised imitation from the fixed dataset $\mathcal{D}$. Section 2.1.3 explained why plain behavior cloning is fragile over long horizons: the student is evaluated on the states that its own errors produce. For cyclic threading the mechanism is concrete. A student that grasps a millimeter too high applies a slightly different torque, the nut advances a little less, and after a few cycles the fingers close on the edge of the nut instead of on its faces. Three measures counter this compounding in the present method, and they are complementary. First, the demonstrations themselves contain recoveries, because the closed-loop teacher was disturbed by impulses during collection; the student thus sees off-nominal states with correct labels, which is the effect DAgger produces by querying the expert on student-visited states, obtained here without an expert that can be queried at arbitrary states. Second, the student predicts action chunks and executes a temporal ensemble of overlapping

chunks, which filters the single-step noise that would otherwise be amplified by the controller. Third, where a queryable expert exists, in simulation, the same flow-matching student can be trained online with the hybrid teacher labeling the states that the student visits; this online variant is described in Section 3.5.5 and compared with the offline variant in Chapter 4. The deployed student is trained offline on $\mathcal{D}$: the dataset is rendered once under visual randomization and reused unchanged by every student and ablation, and the offline flow student sustains ten cycles (Chapter 4). Online DAgger serves as a comparison in simulation.

3.5.2 Flow Matching for Conditional Action Chunks

Flow matching learns a generative model as the solution of an ODE. Let $\boldsymbol{x} \in \mathbb{R}^d$ denote a sample, let $p_0 = \mathcal{N}(\mathbf{0}, I)$ be the source distribution, and let $p_1 = q$ be the data distribution. A time-dependent vector field $\boldsymbol{v} : \mathbb{R}^d \times [0,1] \to \mathbb{R}^d$ defines a flow ψ_λ through $\frac{\mathrm{d}}{\mathrm{d}\lambda}\psi_\lambda(\boldsymbol{x}) = \boldsymbol{v}(\psi_\lambda(\boldsymbol{x}), \lambda)$ with $\psi_0(\boldsymbol{x}) = \boldsymbol{x}$, and the flow transports p_0 to the marginal path $p_\lambda = [\psi_\lambda]_{\#} p_0$. The flow-matching objective regresses a parametric field $\boldsymbol{v}_\theta$ onto a target field $\boldsymbol{u}_\lambda$ that generates a prescribed path p_λ from p_0 to p_1:

$$L_{\mathrm{FM}}(\theta) = \mathbb{E}_{\lambda \sim U[0,1],\, \boldsymbol{x} \sim p_\lambda} \left\| \boldsymbol{v}_\theta(\boldsymbol{x}, \lambda) - \boldsymbol{u}_\lambda(\boldsymbol{x}) \right\|_2^2 . \tag{3.9}$$

This objective cannot be evaluated directly, because the marginal path and its field are defined only implicitly by the data. Conditional flow matching [23] resolves this by constructing the marginal path as a mixture of conditional paths $p_\lambda(\boldsymbol{x} \mid \boldsymbol{x}^1)$, one per data sample $\boldsymbol{x}^1 \sim q$, each with a known conditional field $\boldsymbol{u}_\lambda(\boldsymbol{x} \mid \boldsymbol{x}^1)$. The marginal field is the posterior-weighted average of the conditional fields,

$$\boldsymbol{u}_\lambda(\boldsymbol{x}) = \int \boldsymbol{u}_\lambda(\boldsymbol{x} \mid \boldsymbol{x}^1) \, \frac{p_\lambda(\boldsymbol{x} \mid \boldsymbol{x}^1)\, q(\boldsymbol{x}^1)}{p_\lambda(\boldsymbol{x})} \, \mathrm{d}\boldsymbol{x}^1, \tag{3.10}$$

and the conditional objective

$$L_{\mathrm{CFM}}(\theta) = \mathbb{E}_{\lambda \sim U[0,1],\, \boldsymbol{x}^1 \sim q,\, \boldsymbol{x} \sim p_\lambda(\cdot \mid \boldsymbol{x}^1)} \left\| \boldsymbol{v}_\theta(\boldsymbol{x}, \lambda) - \boldsymbol{u}_\lambda(\boldsymbol{x} \mid \boldsymbol{x}^1) \right\|_2^2 \tag{3.11}$$

has the same gradient with respect to θ as (3.9); the two differ only by a constant that does not depend on θ, which follows by expanding the squares and using (3.10). The conditional path can therefore be chosen for convenience. The optimal-transport path takes the conditional distribution to be Gaussian with mean $\lambda \boldsymbol{x}^1$ and standard deviation $1 - \lambda$ (up to a small floor), so that a sample is the straight-line interpolant

$$\boldsymbol{x}^\lambda = (1-\lambda)\boldsymbol{x}^0 + \lambda \boldsymbol{x}^1, \qquad \boldsymbol{x}^0 \sim p_0, \tag{3.12}$$

and the conditional field along this line is constant, $\boldsymbol{u}_\lambda(\boldsymbol{x}^\lambda \mid \boldsymbol{x}^1) = \boldsymbol{x}^1 - \boldsymbol{x}^0$. Straight conditional paths make the learned marginal field nearly straight as well, which is why few integration steps suffice at sampling time.

The student applies this construction to action chunks conditioned on the observation. At control step t the student observes $\boldsymbol{y}_t = (\boldsymbol{O}_t^{\mathrm{RGBD}}, \boldsymbol{s}_t, p_t, \boldsymbol{e}_t^m)$, consisting of the RGB-D image, the proprioception $\boldsymbol{s}_t$, the normalized task progress p_t, and the process-phase encoding $\boldsymbol{e}_t^m$, and learns the distribution of the executed H-step action chunk. The supervision target is the demonstrated chunk $\boldsymbol{U}_t^D = [\boldsymbol{u}_t, \ldots, \boldsymbol{u}_{t+H-1}] \in \mathbb{R}^{H \times 9}$, the observation is encoded into a condition vector $\boldsymbol{c}_t$ by the network of Section 3.5.3, and the distillation objective is

$$\theta^* = \arg\min_\theta \, \mathbb{E} \left\| \boldsymbol{v}_\theta\big(\boldsymbol{U}_t^\lambda, \lambda \mid \boldsymbol{c}_t\big) - \dot{\boldsymbol{U}}_t^\lambda \right\|_2^2, \qquad \boldsymbol{U}_t^\lambda = (1-\lambda)\boldsymbol{U}_t^0 + \lambda \boldsymbol{U}_t^D, \quad \dot{\boldsymbol{U}}_t^\lambda = \boldsymbol{U}_t^D - \boldsymbol{U}_t^0, \tag{3.13}$$

where the expectation is over $(\boldsymbol{y}_t, \boldsymbol{U}_t^D) \sim \mathcal{D}$, $\boldsymbol{U}_t^0 \sim \mathcal{N}(\mathbf{0}, I)$, and $\lambda \sim U[0,1]$. The learned field transports Gaussian noise to the conditional action distribution $p_\theta(\boldsymbol{U}_t \mid \boldsymbol{y}_t)$. At inference, a chunk is generated by drawing $\boldsymbol{U}^0 \sim \mathcal{N}(\mathbf{0}, I)$ and integrating the ODE with the explicit Euler method on a uniform grid of N steps,

$$\boldsymbol{U}^{\lambda_{n+1}} = \boldsymbol{U}^{\lambda_n} + \frac{1}{N}\, \boldsymbol{v}_\theta\big(\boldsymbol{U}^{\lambda_n}, \lambda_n \mid \boldsymbol{c}_t\big), \qquad \lambda_n = \frac{n}{N}, \quad n = 0, \ldots, N-1, \tag{3.14}$$

with $N = 16$ at deployment. The condition $\boldsymbol{c}_t$ is computed once per chunk, so the image is encoded once and the N velocity evaluations cost only the Transformer of Section 3.5.3.

Three properties distinguish this head from the alternatives that the thesis compares against. Relative to a deterministic regression head, which minimizes a squared error between a single predicted chunk and the label and therefore converges to the conditional mean, the flow head represents multimodal labels; this matters because the teacher's recoveries after impulses are state-dependent in ways the observation does not fully resolve, so that near-identical observations can carry the nominal action in one demonstration and a recovery in another. Relative to a diffusion head, which learns the score of a noised data distribution and samples by a stochastic reverse process with many steps along curved paths, the flow head learns a deterministic ODE along nearly straight paths, trains with a plain regression target without a noise schedule, and samples in sixteen steps. The remaining stochasticity of the flow head comes from the source sample only; a zero-noise sample at the source mean is available as a diagnostic.

3.5.3 Student Architecture as Implemented

Figure 3.4 shows the student. It consists of a vision encoder, a proprioception encoder, a fusion and conditioning stage, and the velocity network.

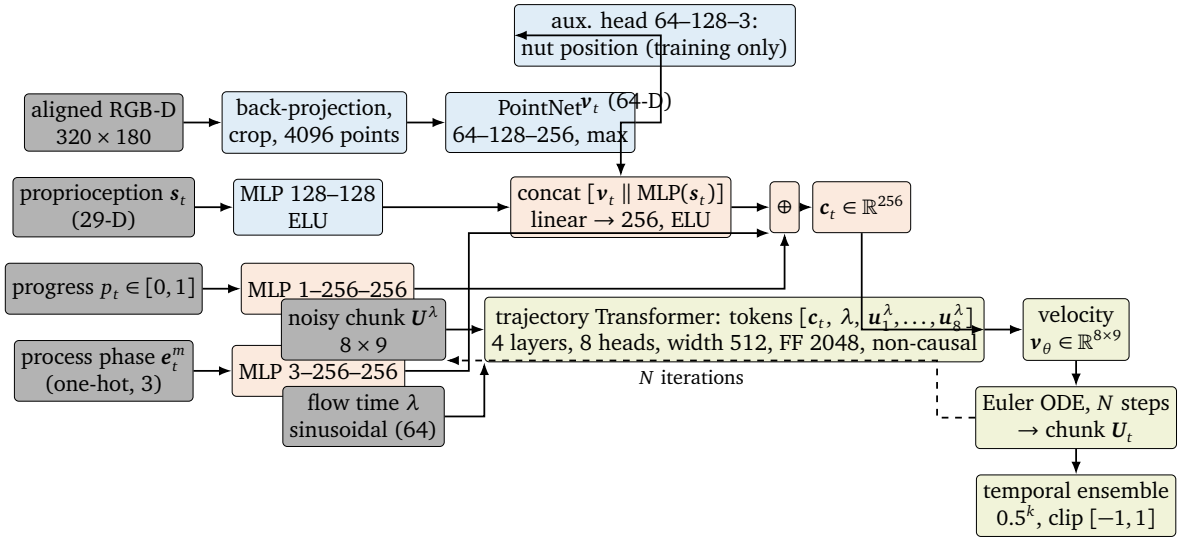


Figure 3.4: Student architecture. The aligned RGB-D observation is back-projected into a point cloud and encoded into a 64-D visual feature; the 29-D proprioception is encoded by an MLP; late fusion concatenates the two with the visual feature first and projects to the 256-D condition, to which learned embeddings of the global trajectory progress and of the process phase are added. A non-causal trajectory Transformer predicts the flow velocity of all eight action tokens given the condition token and a sinusoidal embedding of the flow time. An auxiliary head regresses the nut position from the visual feature during training.

Observation encoding and late fusion. Following DextrAH-RGB [24], the visual and proprioceptive streams are encoded separately and fused by concatenation, with the visual feature placed before the encoded proprioception. The 29-D proprioception $\boldsymbol{s}_t$, consisting of the ten controlled joint positions, the ten joint velocities, and the previous filtered 9-D action, is encoded by a 128–128 MLP with ELU activations. The concatenation of the 64-D visual feature and the 128-D proprioceptive feature is projected by a linear layer with ELU activation to the 256-D condition. In the privileged ablations of Chapter 4 the vision encoder is replaced by the identity and the nut position (3-D) or the nut position with its canonical axis-angle orientation (6-D) is appended to the proprioception instead; in the ablation that removes the hand velocities, the three pinch-joint velocities are dropped and the proprioception becomes 26-D.

Vision encoder. The retained encoder follows DP3 [27]. The depth image is back-projected through the calibrated intrinsics into camera-frame points; points with non-finite depth, depth outside $[0.10, 2.0]$ m, or coordinates outside a fixed crop box that covers the workspace with a 40 mm margin are rejected; the remaining points are centered and scaled by fixed workspace constants and carry the color of their pixel, normalized to $[-1, 1]$. From the valid points, 4096 are sampled, randomly during training and with a fixed priority at evaluation; training fails fast if fewer than 128 valid points remain. A compact PointNet with pointwise layers of width 64, 128, and 256, layer normalization, ReLU activations, and global max pooling, followed by an ELU projection, yields the 64-D visual feature $\boldsymbol{v}_t$. The encoder has about 60 000 parameters and no pretrained backbone; it can be initialized from a pose-regression pretraining on rendered sweeps of a free-moving nut, which Chapter 4 evaluates. The alternative encoders compared in Chapter 4 share the same 64-D interface; two of them are described here. An image-space encoder applies an ImageNet-pretrained ResNet-18 stage-2 trunk to the concatenation of RGB, log depth, a validity mask, and the calibrated camera-frame XYZ of each pixel, followed by one spatial self-attention layer and learned-query pooling; this is the ablation (b) of Chapter 4. An Adapt3R-style encoder [36] extracts dense features with a frozen ResNet-18 layer-1 backbone, localizes 512 of them in 3D through the depth map, expresses the points in the live grasp frame, and pools them by learned-query attention. All encoders consume the same 320×180 RGB-D pair.

Conditioning on progress and phase. Two scalar-sized conditions are added to the 256-D condition through separate learned embeddings, each a 1–256–256 or 3–256–256 MLP with ELU activation. The global trajectory progress $p_t \in [0, 1]$ maps the first timestamp of each demonstration to 0 and the last to 1; at deployment it advances with elapsed time from the start of the trial, normalized by the median demonstration duration stored in the checkpoint, and is clamped to 1 afterwards. It is a monotonic clock for the complete behavior and nothing more: it is not a teacher-stage selector, it is not periodic, and it does not encode the cycle index; it tells the student how far into a nominally timed trial it is, which disambiguates, for instance, the first and the last approach of a trial that otherwise look alike. The process-phase encoding $\boldsymbol{e}_t^m$ is a three-way one-hot over the policy-controlled phases, the release waypoints, and the return; during training it is derived from the recorded phase labels, and during evaluation it is published by the cyclic coordinator from physical events, so that a cycle that takes longer than nominal cannot select the wrong phase. The phase encoding is a synchronization aid: it does not switch networks or action contracts, and the student predicts the transition actions itself.

Velocity network. The velocity field is a non-causal trajectory Transformer. The noisy chunk $\boldsymbol{U}^\lambda$ is split into $H = 8$ action tokens of dimension 9, each embedded linearly into

the 512-D model width; a condition token embeds $\boldsymbol{c}_t$ and a time token embeds the flow time λ through a 64-D sinusoidal encoding. Learned positional embeddings mark the ten token positions; they encode only the offset of an action inside the chunk. Four pre-norm Transformer encoder layers with eight attention heads, a 2048-D GELU feed-forward block, and no dropout apply full self-attention over all tokens, so that every future step may attend to the whole noisy chunk when predicting its velocity, and a linear output layer maps each action token to its 9-D velocity. The model keeps an exponential moving average of its parameters with decay 0.999, and inference uses the averaged weights. An auxiliary head, a 64–128–3 MLP on the visual feature, regresses the nut position in normalized coordinates with unit weight; it regularizes the visual feature toward spatially meaningful content and is discarded at deployment. Training uses the Adam optimizer with learning rate 10^{-4}, batches of 64 windows, 300 epochs, and DextrAH-style RGB augmentation of the stored images, with saturation, contrast, and brightness factors in $[0.5, 1.5]$, hue shifts in $[-0.15, 0.15]$, and motion blur with probability 0.1.

Action chunking and execution. With $H = 8$ and 15 Hz, one chunk covers 0.53 s. Chunk targets begin at the sample paired with the observation (target offset zero), which is the correct alignment for the Cartesian contract because the recorder stores the pre-step state together with the command acting on it. At deployment the student replans at every control step, and the command sent to the controller is a temporal ensemble of all still-valid proposals for the current tick with weights 0.5^k for a proposal of age k; a decay of zero recovers strict first-action execution. The ensemble operates in action space and is distinct from the parameter-space EMA. The executed 9-D action is clipped to $[-1, 1]$ and passes through the unchanged controller path of Section 3.2.3, including the exponential action smoothing. Inference at 15 Hz with 16 Euler steps on an RTX 5090 fits within the control period; the measured latency is TBD.

Normalization. The Cartesian actions are already normalized to $[-1, 1]$ by the action contract, so no action statistics are learned. Proprioceptive inputs are consumed in physical units; the nut position of the auxiliary head is standardized by dataset statistics stored in the checkpoint.

3.5.4 Offline Training

Algorithm 1 summarizes offline distillation. The dataset consists of the simulated hybrid-teacher rollouts, windows of H consecutive commands are built from each episode without crossing episode boundaries, and the velocity field is trained with (3.13) while an EMA copy of the parameters is maintained.

3.5.5 Online DAgger with a Flow-Matching Head

The online variant replaces the fixed dataset by live teacher labels. The simulator is driven by a mixture of teacher and student control with teacher probability β, the hybrid teacher labels every visited state with its 9-D command, and the flow head is trained on the labels as they arrive. Two aspects differ from the recurrent DAgger of DextrAH and of the predecessor thesis. First, the label of a chunked student is a sequence of H future teacher commands, not a single command. The trainer collects eight consecutive live teacher commands, pairs the sequence with the observation at its first step, and uses stride-one overlapping supervision, so

Algorithm 1 Offline flow distillation

Require: hybrid teacher π_{HT}, horizon H

Collect simulated rollouts $\mathcal{D}_{\mathrm{sim}}$ with π_{HT} under random arm impulses and visual domain randomization
Form H-step chunks from $\mathcal{D} = \mathcal{D}_{\mathrm{sim}}$, masking resets and padding terminal chunks
repeat
    Sample $(\boldsymbol{y}_t, \boldsymbol{U}_t^D) \sim \mathcal{D}$, $\boldsymbol{U}_t^0 \sim \mathcal{N}(\mathbf{0}, I)$, $\lambda \sim U[0,1]$
    Encode the point cloud and fuse it with proprioception, progress, and phase to obtain $\boldsymbol{c}_t$
    Update θ by one gradient step on (3.13) plus the auxiliary nut-position loss
    Update EMA parameters $\bar{\theta} \leftarrow \alpha\bar{\theta} + (1-\alpha)\theta$
until convergence
return student velocity field $\boldsymbol{v}_{\bar{\theta}}$

that once the initial horizon is full one chunk completes at every control step; sequences that cross a reset or the scripted handover are discarded. Second, the chunk label is only a valid sample of $p(\boldsymbol{U}_t \mid \boldsymbol{y}_t)$ if the teacher itself produced the states on which the later commands were computed. If the student had controlled some of those steps, the later labels would describe the teacher's reaction to a student-induced future, not the teacher's trajectory from the conditioning observation. Multi-step online supervision therefore fixes $\beta = 1$ during collection: the teacher carries every training state, and the student is exposed to its own state distribution only at evaluation. The RGB-D latent is encoded once per step and detached before it enters the pending windows; the vision encoder is trained by the immediate nut-position auxiliary objective at a reduced learning rate of 10^{-5}, and the delayed flow objective trains the action path from the stored latent with learning rate 10^{-4}. Sampling during online evaluation uses eight Euler steps. Algorithm 2 summarizes the loop.

Algorithm 2 Online DAgger with a conditional flow-matching head

Require: hybrid teacher π_{HT}, student $\boldsymbol{v}_\theta$, horizon H, teacher probability β

for each iteration **do**
    Observe $\boldsymbol{y}_t$ in every environment; query the teacher label $\boldsymbol{u}_t^* = \pi_{\mathrm{HT}}(\boldsymbol{x}_t, m_t)$
    Sample a student chunk from $\boldsymbol{v}_\theta$ with N Euler steps and take its first action $\hat{\boldsymbol{u}}_t$
    Step each environment with $\boldsymbol{u}_t^*$ with probability β, otherwise with $\hat{\boldsymbol{u}}_t$
    Append $\boldsymbol{u}_t^*$ to the pending windows that started at steps $t-H+1, \ldots, t$
    Emit every completed window $(\boldsymbol{y}_{t-H+1}, [\boldsymbol{u}_{t-H+1}^*, \ldots, \boldsymbol{u}_t^*])$ and drop windows that crossed a reset
    Update θ on the emitted windows with (3.13); update the EMA parameters
end for

With a one-step horizon the constraint on β disappears, and the intervention schedule becomes a design choice. The code base implements three schedules for the recurrent baseline and the one-step case: a fixed β, a linear decay from β_{start} to β_{end} over a given number of iterations, and a SafeDAgger rule that hands control to the teacher when the RMSE between the deterministic teacher and student commands exceeds a takeover threshold of 0.10 and returns it when the error falls below a release threshold of 0.075; the hysteresis produces coherent corrective segments rather than per-step flicker. Which schedule allows a cyclic student to learn both the turn and the transitions is an experimental question that Chapter 4 answers.

3.5.6 Baseline Students

Three deterministic students serve as baselines under the same observation and action contracts. The behavior-cloning baseline consumes an eight-frame history of measured joint positions and velocities and predicts two future commands with a 512–512–512 MLP with batch normalization; it acts at the physics rate and follows a reference cursor, which makes it a plumbing and local-imitation diagnostic rather than a candidate for deployment. The recurrent DAgger baseline follows the DextrAH distillation recipe: a 512-unit layer-normalized LSTM followed by a 512–512–256 ELU MLP and a Gaussian head, trained by deterministic action regression on the teacher command, with the student's deterministic mean as the rollout command. The Transformer regression baseline is a deterministic variant of ACT [56] without the variational objective: four layers, eight heads, 512-D hidden width, 2048-D feedforward layers, and eight learned action queries that predict an eight-step chunk. The last two baselines isolate the contribution of the generative head from that of chunking and of the backbone, and they are the deterministic regression comparisons of RQ2.

3.6 Evaluation Protocol

3.6.1 Evaluation Ladder

A failed closed-loop rollout can originate in the supervised fit, in the command conversion and controller, or in the distribution shift between teacher and student states. The three evaluation modes in Figure 3.5 separate these causes, and each mode makes a stronger claim than the previous one.

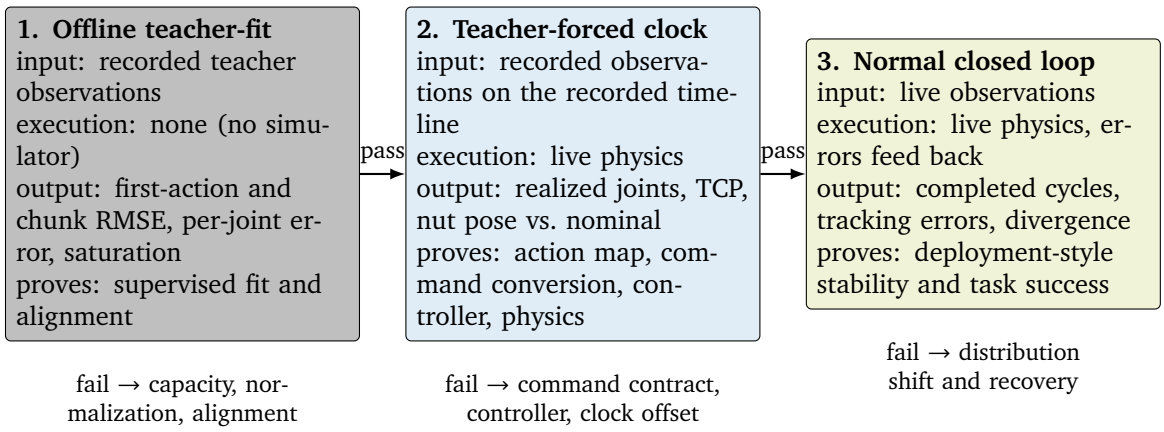


Figure 3.5: Evaluation ladder. Offline teacher-fit feeds recorded teacher observations to the student and compares predicted chunks with recorded commands without a simulator. Teacher-forced clock executes the student's commands in live physics while the student receives the recorded teacher observations on the recorded timeline, removing observation drift. Normal closed loop feeds live observations back to the student and is the only mode that measures compounding error.

In the offline teacher-fit mode, every recorded teacher observation of a demonstration is fed to the student, and the sampled chunk is compared with the recorded command chunk; the reported quantities are the RMSE of the first action and of the full chunk, per-joint errors over time, and target saturation. Passing shows that the checkpoint can represent the observation-to-action mapping under the chosen alignment; it establishes neither controller correctness nor closed-loop success. In the teacher-forced clock mode, the student's pre-

dicted commands are executed in live physics, but the student receives the recorded teacher observations selected on the recorded playback timeline, and for progress-conditioned checkpoints the same timeline supplies p_t; the realized joints, tool center point, and nut pose are compared with the recording. Passing shows that the learned action map, the command conversion, the controller, and the physics reproduce the nominal behavior once closed-loop observation drift is removed. Only the normal closed-loop mode, in which the student consumes live observations and its errors feed back into later observations, measures task success and compounding error; it is the mode of all results that claim success. The two diagnostic modes are used in that order to localize a failure before expensive retraining: a bad offline fit points to capacity, normalization, or alignment; a good fit with bad teacher-forced execution points to the command contract; and good teacher-forced execution with a bad closed loop points to distribution shift and recovery, which is the regime that motivates online labels or disturbance augmentation.

3.6.2 Metrics

Each condition is evaluated over $N_{\mathrm{tr}} = 10$ trials with up to ten cycles per trial, where each cycle consists of a 60° clockwise turn followed by release and return. Three quantities are reported. The mean completed cycle count $\bar{C} = N_{\mathrm{tr}}^{-1} \sum_i C_i$ counts, for trial i, the cycles $C_i \in \{0, \ldots, 10\}$ completed before failure or timeout; incomplete cycles are excluded, and higher is better. The cyclic tracking error is computed for every completed cycle k of trial i as $e_{i,k} = \Delta\varphi_{i,k} - 60°$, where $\Delta\varphi_{i,k} = \varphi_{i,k} - \varphi_{i,k-1}$ is the clockwise nut rotation; over all $M = \sum_i C_i$ completed cycles the mean $\mu_{\mathrm{cyc}} = M^{-1} \sum_{i,k} e_{i,k}$ and standard deviation σ_{cyc} are reported, where negative values indicate lag and positive values overshoot, and a mean closer to zero with a smaller deviation is better; if $M = 0$ both statistics are undefined. The ten-cycle tracking error is computed over the set $I_{10} = \{i : C_i = 10\}$ of trials that complete all cycles as $e_{10,i} = \sum_{k=1}^{10} \Delta\varphi_{i,k} - 600°$, reported as mean μ_{10} and standard deviation over I_{10}; it is undefined when no trial completes. All errors are in degrees. In simulation the nut angle is read from the simulator; on hardware it is read from an angular scale mounted under the bolt (Chapter 5).

With every component of the method and its evaluation defined, Chapter 4 applies the protocol in simulation to answer RQ1 and RQ2, and Chapter 5 applies it on the physical system to answer RQ3.

Chapter 4

Experiments in Simulation

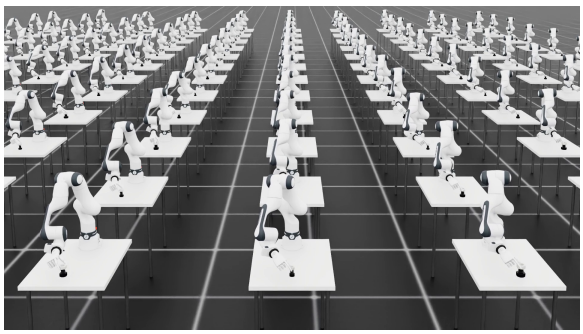
This chapter tests the method of Chapter 3 in simulation, where every component can be varied under matched conditions. It answers RQ1 and RQ2 of Section 1.4; RQ3, the zero-shot transfer, needs the physical system and is evaluated in Chapter 5. Section 4.1 fixes the setup that all experiments share. Section 4.2 shows that the privileged teacher and the hybrid rollout produce reliable multi-cycle demonstrations, the precondition for every student. Section 4.3 reports the main ablations that answer RQ1 and RQ2. Section 4.4 reports the design ablations that fixed the action contract, the conditioning, and the encoder of the student, and Section 4.5 compares offline with online distillation. Section 4.6 lists every ablation that the code base supports and marks which of them have results, and Section 4.7 summarizes the chapter. Numbers that are not yet available are marked TBD.

4.1 Experimental Setup

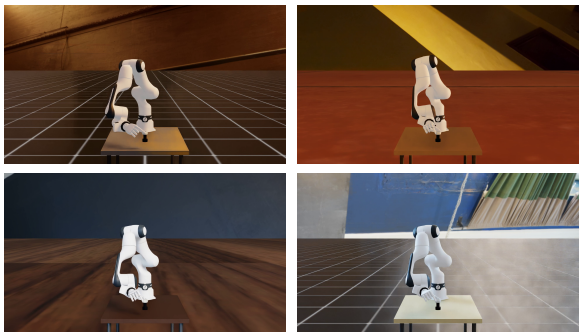
All simulation experiments run the system of Section 3.2 in Isaac Sim 5.1 with physics at 120 Hz and control at 15 Hz on a single NVIDIA GeForce RTX 5090. Each teacher is trained with 512 parallel environments for 200 epochs, which takes about ten hours. The student is trained offline for 300 epochs in about 30 minutes on the dataset of Section 3.4.5: 20 successful ten-cycle demonstrations of the M24 hexagonal nut, collected from 32 environments with 30 random arm impulses each, 19 486 frames in total. The demonstrations are rendered under lighting and texture randomization of the dome light, the tabletop, the nut and bolt textures, the floor texture, and the image background in the style of DextrAH-RGB [24]. Figure 4.1 shows the parallel training scenes and the randomized student views. Unless stated otherwise, students use the Cartesian action contract with $H = 8$, sixteen Euler steps, a temporal-ensemble decay of 0.5, EMA weights, and the DP3-style encoder. Evaluation follows Section 3.6: ten trials of ten cycles on the M24 nut in a nominal scene without physical domain randomization.

4.2 Teacher Training

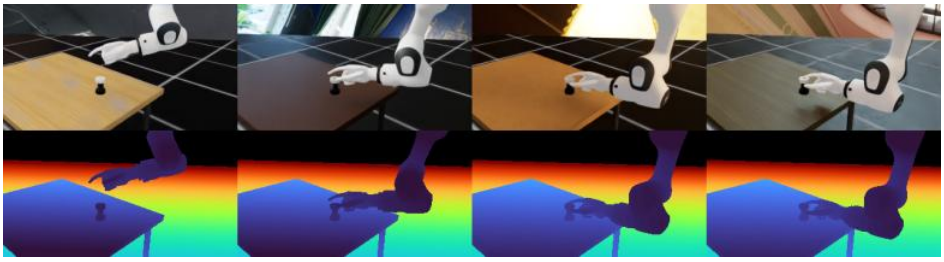
Every demonstration comes from the hybrid teacher, so the teacher has to turn the nut reliably and hand over cleanly to the scripted release and return. The teacher result is also the experimental test of difficulty (b) of Section 1.2: whether a learned policy acquires the pinch grasp and keeps it under the turning load through the commanded hand and arm degrees of freedom. With the simplified thread model and the three-stage reward of Section 3.3, the



(a) Parallel teacher-training workcells.



(b) Lighting and texture randomization.



(c) Student RGB (top) and depth (bottom) observations in simulation during approach, grasp, rotation, and release.

Figure 4.1: Simulation scenes. Teacher policies are trained in hundreds of parallel workcells rendered without visual randomization; demonstrations for the student are rendered with randomized dome lighting, floor, tabletop, and object textures, and the student sees aligned RGB and metric depth of the resulting scene.

threading expert learns the single 60° turn from the engaged nut. On an earlier humanoid-arm platform, the recipe with the ±30° initial-yaw curriculum reached a population success rate of 83.6 % at the end of training, with a peak of 90.0 %. On the final platform, the retained M24 teacher, trained for 200 epochs with the curriculum and the axial-height randomization, completed ten consecutive cycles in all 32 hybrid-teacher candidate environments without a single transition failure. This is the property that matters for demonstration generation. The single-turn success rate of this teacher at the final curriculum level is TBD. Table 4.1 summarizes the teacher results. It shows that the teacher on the final platform meets the requirement of the student dataset, error-free transitions over ten cycles, while its single-turn success rates are still pending.

Table 4.1: Teacher training results. Success is the population success rate at the end of training; the hybrid-teacher column counts candidate environments that completed the required cycles with zero transition failures. The earlier platform is the humanoid arm on which the project started; the final platform is described in Section 5.1.

Teacher	Platform	Success	Peak success	Hybrid rollout
Threading, ±30° yaw curriculum	earlier humanoid arm	83.6 %	90.0 %	–
Threading, M24, yaw and depth randomization	final	TBD	TBD	32/32 ten-cycle
Threading, M24/M30/M36 one-hot	final	TBD	TBD	TBD

The pickup expert of the full-task variant (Section 3.4.2) learns to approach, pinch, and lift a loose nut with the same recipe. It reached 95.3 % grasp and 94.5 % lift success on the earlier platform and 99.8 % pinch success on the final platform. Seating the nut on the bolt is outside the scope of this thesis (Section 3.2.2), so all threading results start from an engaged nut. All reported teachers use the Factory controller gains of Section 3.2.3, and the physical controller is tuned toward these gains (Chapter 5).

4.3 Student Ablations in Simulation

The main ablation study answers RQ1 and RQ2 of Section 1.4. Five students are trained on the simulated dataset of Section 3.4.5 and evaluated under matched budgets, and each ablation changes exactly one component of the full method. Ablation (a) addresses RQ1: it tests whether one student trained on multi-cycle data outperforms a single-cycle policy that is invoked repeatedly. Ablations (b) to (d) address RQ2: (b) replaces the point-cloud encoder with an image-space encoder, and (c) and (d) replace the flow-matching head with deterministic regression on a recurrent and on a Transformer backbone. RQ3 is evaluated on hardware in Chapter 5.

(a) *Single-cycle student.* The student is trained on demonstrations of one cycle and is invoked ten times in a trial; its temporal context is reset after every cycle.

(b) *ResNet visual encoder.* The point-cloud encoder is replaced by the one-branch image-space encoder of Section 3.5.3, an ImageNet-pretrained ResNet-18 stage-2 trunk that fuses RGB, log depth, a validity mask, and the calibrated per-pixel XYZ coordinates into the same 64-D feature. Policy, objective, data, and budget are unchanged.

(c) *Point-cloud encoder with LSTM regression.* The flow-matching Transformer is replaced by the DextrAH-RGB-style recurrent backbone of Section 3.5.6, a layer-normalized 512-unit LSTM followed by a 512–512–256 ELU MLP, trained by deterministic action regression on the same data.

(d) *Point-cloud encoder with Transformer regression.* The deterministic ACT-style Transformer of Section 3.5.6 predicts the eight-step chunk from eight learned action queries.

(e) *Full method.* One flow-matching student with the point-cloud encoder is trained on the complete multi-cycle simulated demonstrations.

Table 4.2 reports the results on the M24 hexagonal nut. The full method (e) completes all ten cycles in every trial, with a cyclic tracking error of $-1.77 \pm 1.02°$ and a ten-cycle error of $-17.7 \pm 3.23°$. The student thus lags the $60°$ target by less than two degrees per cycle, and the lag accumulates over the trial without its spread drifting. Each ablation removes one property on which this result may depend. The single-cycle student (a) never sees the axial advance of the nut: in a single-cycle demonstration the nut is always grasped at its reset height, whereas over ten cycles it rises by one sixth of a pitch per cycle, and the multi-cycle student has to read this height from the point cloud (RQ1). The ResNet student (b) keeps everything but the representation. The regression students (c) and (d) keep the data and, for (d), also the backbone, the chunk length, and the temporal ensembling, so that their difference to (e) isolates the generative objective (RQ2). Their results on the simulated dataset are TBD.

For the encoder part of RQ2, a representation probe complements the closed-loop ablation (b). One shared pose decoder is trained on the frozen 64-D feature of each encoder to predict the nut position and orientation. A re-grasp needs exactly the orientation of the nut faces relative to the fingers, so the probe tests whether a representation carries the quantity on which the closed loop depends. The probe errors of the point-cloud and the ResNet features are TBD.

4.4 Design Ablations of the Student

The main ablations compare the final student with alternatives at the level of whole components. The design of the student itself was fixed earlier by a sequence of controlled ablations,

Table 4.2: Simulation ablations on the M24 hexagonal nut over ten trials. Each defined entry reports mean ± standard deviation. Completed cycles are averaged across trials, cyclic-error statistics use completed cycles only, and the ten-cycle error compares the accumulated rotation with the 600° target; it is reported as "–" when no trial completes all ten cycles. Errors are signed in degrees: negative denotes lag and positive overshoot.

Ablation	Completed cycles ↑	Cyclic error → 0	Ten-cycle error → 0
(a) single-cycle student	TBD	TBD	TBD
(b) ResNet encoder	TBD	TBD	TBD
(c) LSTM regression	TBD	TBD	TBD
(d) Transformer regression	TBD	TBD	TBD
(e) full method	10.0 ± 0.0	-1.77 ± 1.02	-17.7 ± 3.23

which this section reports because they explain why each component is present. The action-contract comparison uses pickup demonstrations; the remaining design ablations use six-cycle sequential-threading data. Except where a subsection varies them, all runs use the privileged 6-D nut-pose observation or the RGB-D observation as stated, 20 impulse-augmented demonstrations, 300 epochs, the Cartesian contract, $H = 8$, and the student-owned cyclic evaluation with a six-cycle requirement at a fixed seed. The results are matched single rollouts, not success rates.

4.4.1 Action Contract

The student imitates the teacher's bounded, bolt-relative Cartesian action because a joint-space contract fails in closed loop despite an equally good offline fit. Table 4.3 compares the two contracts on pickup demonstrations with the privileged 6-D observation. Materialized in the native Cartesian contract of Section 3.2.3, the demonstrations produced a flow student with 10/10 grasps and lifts, 5/10 terminal successes on the handoff-only task, and 8/10 qualified handoffs, 7/10 engagement entries, and 2/10 held engagements on the stricter engagement task. Materialized in the 17-D joint-PD contract, the same demonstrations produced a student that fitted its labels equally well offline (training flow loss 0.03, teacher-fit chunk RMSE 0.040, correlation 0.998) and grasped and lifted in every trial, but reached the handoff waypoint in none. Neither terminal-hold padding nor a residual variant that predicts arm-position residuals reached the handoff waypoint either. A good offline fit therefore turns into closed-loop task progress only when the student imitates the action that the teacher itself produced, and all later students use the Cartesian contract.

Table 4.3: Action-contract ablation of the offline flow student on pickup demonstrations (privileged 6-D observation, single seed). G0 is the handoff waypoint above the bolt tip and G1 the engagement goal.

Contract	Task	Grasp	Lift	G0	G1 held
17-D joint-PD, absolute	engagement	10/10	10/10	0/10	0/10
17-D joint-PD, terminal padding	engagement	10/10	10/10	0/10	0/10
10-D joint-PD, residual	engagement	0/10	0/10	0/10	0/10
9-D Cartesian (native)	handoff only	10/10	10/10	5/10	–
9-D Cartesian (native)	engagement	10/10	10/10	8/10	2/10

4.4.2 Progress and Phase Conditioning

The global progress scalar lets the student finish all cycles, and the process-phase encoding keeps it synchronized with the physical cycle. On six-cycle sequential-threading demonstrations with policy-only impulses, a state-conditioned flow student completed four of the six cycles. With impulses in every phase segment and the global trajectory progress p_t added to the condition, the student completed all six required cycles in 735 control ticks with zero transition failures. Its release times, however, drifted increasingly behind the teacher, by +8.6 s at the sixth release, and the fifth cycle took twice as long as the others. Adding the three-way process-phase encoding reduced the completion time to 609 ticks and equalized the cycle durations to 6.7 to 7.1 s. It also reduced the release-time errors against the teacher to within ±0.5 s, the controlled-joint RMSE against the nominal trajectory from 11.5° to 4.7°, the TCP position RMSE from 75 mm to 28 mm, and the final cumulative nut-turn error from −60° to −8°. The offline teacher fit of the phase-conditioned checkpoint was marginally worse than that of the progress-only checkpoint (first-action RMSE 0.0046 against 0.0044), so a better fit does not explain the closed-loop improvement. The phase encoding acts instead as a synchronization signal that keeps the student's internal clock aligned with physical events when the cycle duration deviates from nominal. All later students use both conditions. Table 4.4 collects the numbers.

Table 4.4: Conditioning ablation on six-cycle sequential threading (privileged 6-D observation, matched single rollouts at one seed). RMSE values are against the nominal teacher trajectory in absolute time; the final turn error is the cumulative nut-rotation error after the last cycle.

Condition	Cycles	Ticks	Joint RMSE	Final turn error
state only, policy-only impulses	4/6	776	14.9°	–
state + progress p_t, all-phase impulses	6/6	735	11.5°	−60.3°
state + progress + phase $\boldsymbol{e}_t^m$	6/6	609	4.7°	−8.2°

4.4.3 Vision Encoder and Pretraining

The visual students match the privileged student, and the retained DP3 encoder owes its accuracy to pretraining on rendered nut sweeps. The first RGB-D student used the two-branch cross-attention encoder with an auxiliary nut-position head and completed all six cycles in 617 ticks, matching the privileged student; a recollection with explicit success markers and retraining reproduced the result in 627 ticks. The retained DP3 encoder is pretrained on rendered sweeps of a free-moving nut from the calibrated camera and reaches a held-out position error of 5.9 mm and a hexagon-symmetry-aware rotation error of 46°. Fine-tuning on hybrid-teacher images with phase-balanced sampling reduces these errors to 1.9 mm and 3.8° on held-out teacher episodes, and to 2.8 mm and 6.6° on images from a student rollout. The ten-cycle DP3 student of Table 4.2 was initialized from this sweep-pretrained encoder with all encoder parameters trainable. It completed all ten cycles in 1072 ticks with a maximum turn of 60.3° and a mean absolute retained-turn error of 3.1°. Its release times lag those of its source demonstration by 2.7 s on average and by 7.4 s at the tenth cycle, a timing drift that does not affect the completion of the cycles. Table 4.5 prepares a closed-loop comparison of the DP3, Adapt3R-style, ResNet, two-branch, and depth-only encoders under identical data and budget, which extends ablation (b); its results are TBD.

Table 4.5: Vision-encoder ablation of the flow student on the ten-cycle M24 task (ten trials). The DP3 row is the full method; the remaining rows are placeholders.

Encoder	Completed cycles ↑	Cyclic error → 0	Ten-cycle error → 0
point cloud, DP3-style (RGB + XYZ)	10.0 ± 0.0	-1.77 ± 1.02	-17.7 ± 3.23
ResNet-18 early fusion (b)	TBD	TBD	TBD
Adapt3R-style (frozen ResNet + 3D attention)	TBD	TBD	TBD
Two-branch cross-attention RGB-D	TBD	TBD	TBD
Depth-only CNN	TBD	TBD	TBD

4.4.4 Privileged Observations, Horizon, Integration Steps, and Inputs

The privileged observation modes serve as upper bounds and as diagnostics. With the nut position (3-D) or pose (6-D) appended to the proprioception instead of the visual feature, the same flow student is trained and evaluated without rendering. On six-cycle sequential threading, the privileged 6-D student and the RGB-D student both completed six cycles at matched settings, which shows that the visual feature carries the information the task needs. The remaining sweeps over the action horizon H, the number of Euler steps N, the temporal-ensemble decay, the hand-velocity input, and the terminal-hold padding are implemented as command-line options of the offline trainer and evaluator but have not been run as controlled studies. Table 4.6 gives the skeleton; once filled, it will show how sensitive the reference student is to each of these choices and what inference latency each number of Euler steps costs.

Table 4.6: Student hyperparameter sweeps on the ten-cycle M24 task (ten trials). All cells are placeholders except the reference row.

Sweep	Value	Completed cycles ↑	Cyclic error → 0	Latency
reference	$H = 8$, $N = 16$, decay 0.5	10.0 ± 0.0	-1.77 ± 1.02	TBD
horizon H	1 / 4 / 16	TBD	TBD	TBD
Euler steps N	1 / 4 / 8 / 32	TBD	TBD	TBD
ensemble decay	0 / 0.8	TBD	TBD	–
hand velocities omitted	26-D proprioception	TBD	TBD	–
privileged 3-D / 6-D nut pose	no vision	TBD	TBD	–

4.4.5 Nut-Size Conditioning, Domain Randomization, and Force Observations

Three further ablations are implemented but not yet qualified in simulation. First, the multi-size teacher conditions on a one-hot code of M24, M30, and M36 with round-robin assignment and per-size meshes, keypoints, and pitches. The student is trained on hexagonal nuts of all three sizes without an explicit size input and must infer the size from the point cloud. Its per-size simulation results are TBD; the hardware results across sizes are reported in Chapter 5. Second, physical domain randomization of mass, friction, controller gains, and reset pose is enabled during teacher training and disabled during demonstration collection and evaluation; its effect on the student when enabled at evaluation time is TBD. Third, hardware-equivalent force observations are implemented as observation channels for both teacher and student: the thumb and index flexion forces projected onto the pad normals, and the seven arm joint reaction torques normalized by the joint torque limits of the arm. They are not yet part of any trained contract, and their effect is TBD. Table 4.7 lists the skeleton, which will show how far the student tolerates size variation, physical perturbation, and

additional force input.

Table 4.7: Robustness ablations in simulation (ten trials of ten cycles). All cells are placeholders.

Ablation	Condition	Completed cycles ↑	Cyclic error → 0	Ten-cycle error → 0
nut size	M24 / M30 / M36	TBD	TBD	TBD
physical DR at evaluation	off / on	TBD	TBD	TBD
force observations	none / fingertip / fingertip + torque	TBD	TBD	TBD
axial-height curriculum	20 mm / TBD	TBD	TBD	TBD

4.5 Offline versus Online Distillation

Offline flow distillation produces students that complete every required cycle, whereas the online DAgger students complete at most two. The comparison uses online DAgger with the recurrent LSTM student on cyclic threading from privileged 6-D observations under the Cartesian contract, with the hybrid teacher as the label source and a ten-cycle cap per episode. The intervention schedule determines which part of the cycle the online student learns. Under full teacher control ($\beta = 1$) it learns the threading turn. Under full student control ($\beta = 0$) with teacher labels it learns the turn and the complete release-and-return transition. With an intervention probability that decays linearly from full teacher to full student control, it learns both and completes two uninterrupted student-only turn-release-return cycles. The offline flow student completes all six cycles of the conditioning study with privileged observations and all ten cycles from RGB-D. Table 4.8 summarizes the comparison.

Table 4.8: Offline and online distillation on cyclic threading (privileged 6-D observation unless stated, Cartesian contract). Cycles are student-only completed cycles of the best retained checkpoint in one evaluation rollout; the offline rows are matched single rollouts from Section 4.4.

Student	Intervention	Completed cycles	Learned behavior
online DAgger LSTM	$\beta = 1$	0	threading turn
online DAgger LSTM	$\beta = 0$	1	turn and transition
online DAgger LSTM	β decayed $1 \rightarrow 0$	2	turn and transition, two cycles
online DAgger LSTM	SafeDAgger 0.10/0.075	0	approach
offline flow, state + progress + phase	–	6/6	complete cycle, Table 4.4
offline flow, RGB-D	–	10/10	complete cycle, Table 4.2

The evaluation ladder of Section 3.6.1 shows that the offline fit does not explain this difference. Table 4.9 places several offline students on its first rung next to their closed-loop outcome. Every listed checkpoint reproduces its training labels with first-action RMSE below 0.04 in normalized action units and correlation above 0.99, and the online LSTM students likewise fit their labels and execute them correctly in teacher-forced mode. The offline fit is therefore a necessary diagnostic and not a predictor of task success; the closed-loop outcome follows the contract, the conditioning, and the data. The retained student combines the three properties that the closed loop rewards: the Cartesian contract, the progress and phase conditions, and demonstrations that contain the teacher's recoveries from disturbances in every phase. Together with chunked prediction and temporal ensembling, these properties plausibly let it recover from its own deviations under its own observations, which is the regime in which the online recurrent students fall behind.

Table 4.9: Offline teacher-fit against closed-loop outcome for offline flow students (EMA weights, sixteen Euler steps, one training episode). RMSE values are in the student's action units; the joint-PD row is in physical units, the Cartesian rows in normalized action units.

Student	Data	First-action RMSE	Chunk RMSE	Correlation	Closed loop
joint-PD, 6-D nut pose	pickup, 20 episodes	0.035	0.040	0.998	0/10 handoff
Cartesian, 6-D nut pose, state only	6-cycle threading, policy impulses	0.0050	0.0110	0.9996	4/6 cycles
Cartesian, 6-D nut pose, + progress	6-cycle, all-phase impulses	0.0044	0.0122	0.9997	6/6 cycles
Cartesian, 6-D nut pose, + progress + phase	6-cycle, all-phase impulses	0.0046	0.0129	0.9997	6/6 cycles
Cartesian, RGB-D two-branch, + aux. head	6-cycle, all-phase impulses	0.0044	0.0055	0.9997	6/6 cycles
Cartesian, DP3 from sweep pretraining	10-cycle, all-phase impulses	0.019	0.019	0.994	10/10 cycles

4.6 Ablation Inventory

Table 4.10 lists every ablation that the code base exposes as a configuration, with the table or section that reports its result or the placeholder that awaits one. The inventory follows the three ablation axes of the student, namely the vision encoder, the action head, and the observation and action contract, and adds the teacher, data, and evaluation options around them. It shows which design choices are backed by a result and which remain open.

Table 4.10: Inventory of ablations supported by the code base. "Result" names the table or section with a result; TBD marks an ablation whose result is pending.

Axis	Ablation	Options	Result
student	behavior cloning (joint history MLP)	diagnostic only	Section 3.5.6
student	online DAgger LSTM	β fixed, decayed, SafeDAgger	Table 4.8
student	offline flow matching	Transformer, MLP backbone	Table 4.2
student	online DAgger flow matching	$\beta = 1$, $H = 8$	TBD
student	regression heads	LSTM (c), Transformer (d)	Table 4.2
vision	encoder	DP3, Adapt3R, ResNet (b), two-branch, depth-only	Tables 4.2, 4.5
vision	DP3 initialization	scratch, sweep pretraining	Section 4.4.3
vision	depth corruption in pretraining	none, Gaussian, realistic	TBD
observation	privileged vs visual	nut XYZ, nut 6-D, RGB-D	Section 4.4.4
observation	hand velocities	included, omitted	TBD
observation	force channels	fingertip force, joint torque	TBD
observation	phase encoding	none, process phase	Table 4.4
observation	progress scalar	none, global progress	Table 4.4
action	contract	9-D Cartesian, 17-D joint-PD, residual	Table 4.3
action	horizon H, steps N, ensemble decay	sweep	TBD
teacher	intervention schedule	see student	Table 4.8
teacher	nut-size conditioning	M24, M30, M36	Chapter 5, TBD
data	disturbance augmentation	none, policy-only, all phases	Table 4.4
data	visual domain randomization	on, off	TBD
data	physical domain randomization	on, off	TBD
evaluation	ladder	offline fit, teacher-forced, closed loop	Section 4.5

4.7 Summary

In simulation, the hybrid teacher produces error-free ten-cycle demonstrations, and the full method trained on them completes all ten cycles of the M24 task in every trial with a cyclic error of $-1.77 \pm 1.02°$. The design ablations attribute this result to the Cartesian contract, the progress and phase conditions, and demonstrations with recoveries from disturbances in every phase, and they show that the offline fit alone does not predict it. Whether the multi-cycle data (RQ1) and the point-cloud encoder and flow-matching head (RQ2) are each necessary will be settled by the reruns of ablations (a) to (d) on the simulated dataset. Simulation cannot show whether the student bridges the gap in contact and perception that the

simplified thread model and the rendered images leave open. Chapter 5 therefore deploys the same student, trained on simulated data only, on the physical system to answer RQ3.

Chapter 5

Experiments on Real Hardware

Chapter 4 showed that the distilled student sustains consecutive threading cycles in simulation. This chapter asks whether that behavior survives the gap between the simplified simulation and the physical workcell, which is RQ3 of Section 1.4: can a student trained on simulated demonstrations only transfer zero-shot to the physical system? No real-world data enters training, and nothing is trained on the robot. Section 5.1 introduces the physical system, including the robot, the camera and its calibration, the controller, the measurement of the nut angle, and the test objects. Section 5.2 describes the zero-shot transfer protocol. Section 5.3 reports the closed-loop results on the M24 training nut, Section 5.4 on unseen nut geometries, and Section 5.5 under long-horizon stress.

5.1 Hardware Setup

The physical workcell, shown in Figure 5.1, realizes the generic platform of Section 3.2.1 with specific products. The arm is a seven-DoF Franka Research 3 (FR3). An Inspire RH56DFX right hand is mounted on its flange through a 10 mm adapter. The kinematic structure of the hand, the three commanded pinch degrees of freedom, and the bench geometry are those of Section 3.2.1.

The simulated system is a model of this workcell. The transform of the adapter flange was measured and reproduced in the simulated asset. The hand asset was regenerated from the manufacturer's URDF, so that joint limits, link inertias, and the mechanical coupling of the finger joints match the physical hand. The FR3 is the arm that replaced the earlier humanoid-arm platform of Section 3.2.1.

A fixed Intel RealSense D415 on a stand provides the RGB-D observation at 640×480. As in simulation, the stream is center-cropped and downscaled to the 320×180 policy input (Figure 5.3). The pose of the camera relative to the robot base was measured by an eye-to-hand calibration, with a translation fit RMSE of 27 mm and a rotation fit RMSE of 3.5° over the calibration poses. The calibrated optical pose and the intrinsics are the ones used to render the simulated camera and to back-project the point cloud, so the student sees the same geometry in both domains.

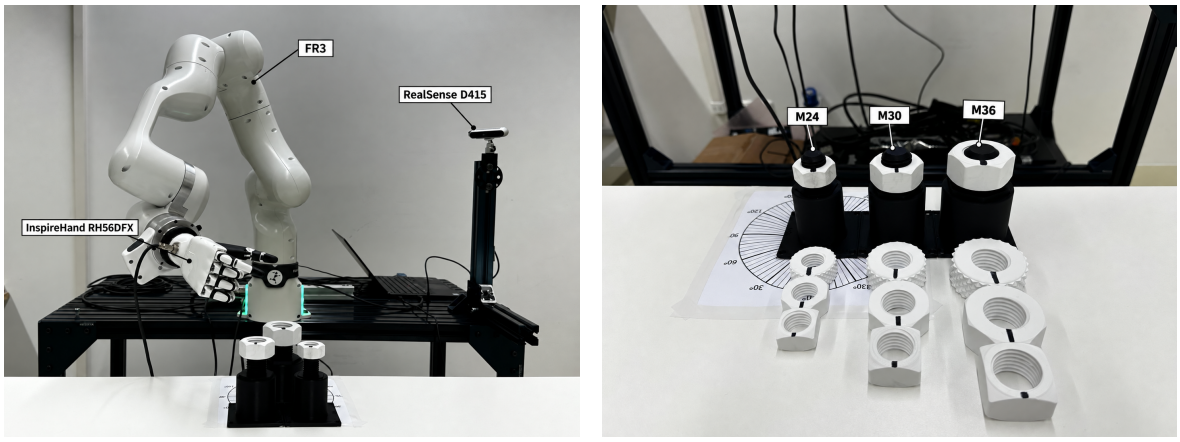
Robot commands use the same Cartesian impedance law as in simulation, so the student's bounded action reaches the arm through the shared action path of Section 3.4.4. The deployment controller is the libfranka Cartesian-impedance example controller. It applies $\tau = J^{\mathrm{T}}(Ke - D\dot{x})$ with a quaternion-vector orientation error and the robot's own gravity compensation; its nullspace term is described below. The controller runs at the 1 kHz rate of the Franka control interface, while the student commands targets at 15 Hz. The hand is position-controlled through its serial interface. The gains of the deployment controller and

the mapping of the student's bounded action to the controller target are TBD.

Controller families. The simulated Factory controller and the physical libfranka-style controller share the task-space law of (2.3) but differ in the nullspace projector. The Factory controller uses the undamped, dynamically consistent torque projector built from the joint-space inertia. The libfranka example instead uses a damped kinematic projector built from a regularized pseudoinverse of the Jacobian transpose with regularization 0.2, and it adds the robot's Coriolis term, which the simulator supplies in the matching simulated backend. Both backends are implemented in the simulation with selectable gains, so a teacher can be trained under either. The reported teachers use the Factory gains, and the physical controller is tuned toward them rather than the other way around. The student's commands therefore meet the stiffness under which the teacher acquired its contact behavior.

Trajectory export. For open-loop validation of the action path, every simulated demonstration can be exported as a command-only deployment artifact. The simulator recording is authoritative for the geometric joint path. Some scripted phases, however, change the joint position kinematically while writing zero articulation velocity, so the export does not use recorded velocities. It fits one twice continuously differentiable quintic curve through the recorded positions and derives position, velocity, and acceleration from it, with deployment limits on joint positions and velocities applied explicitly. Rows with a valid controller command also retain the recorded reverse-engineered 17-D impedance command of Section 3.2.3, so a joint-space replay is possible on controllers that do not expose a Cartesian interface. On the FR3 the Cartesian targets are executed directly through the libfranka-style controller, which keeps open-loop playback and student execution on the same path.

Angle measurement and test objects. The metrics of Section 3.6.2 require the nut angle. It is read from an angular scale printed on a sheet under the bolt pedestal (Figure 5.2), and the unwrapped angle is recorded at the end of every goal-reaching phase. The test-object family, shown on the right of Figure 5.1, consists of M24, M30, and M36 nuts in four profiles each, hexagonal, octagonal, rectangular, and round, with matching bolts and holders. Teacher and student training use the hexagonal profiles only. The other three profiles are held out for the generalization study of Section 5.4.



(a) Physical workcell. **(b)** Test-object family.

Figure 5.1: Hardware setup. Left: the FR3 arm with the Inspire RH56DFX hand, the bolts on their holders in front of the robot, and the fixed RealSense D415 on its stand. Right: M24, M30, and M36 bolts with the hexagonal, octagonal, rectangular, and round nuts of each size; only the hexagonal nuts are used for training.

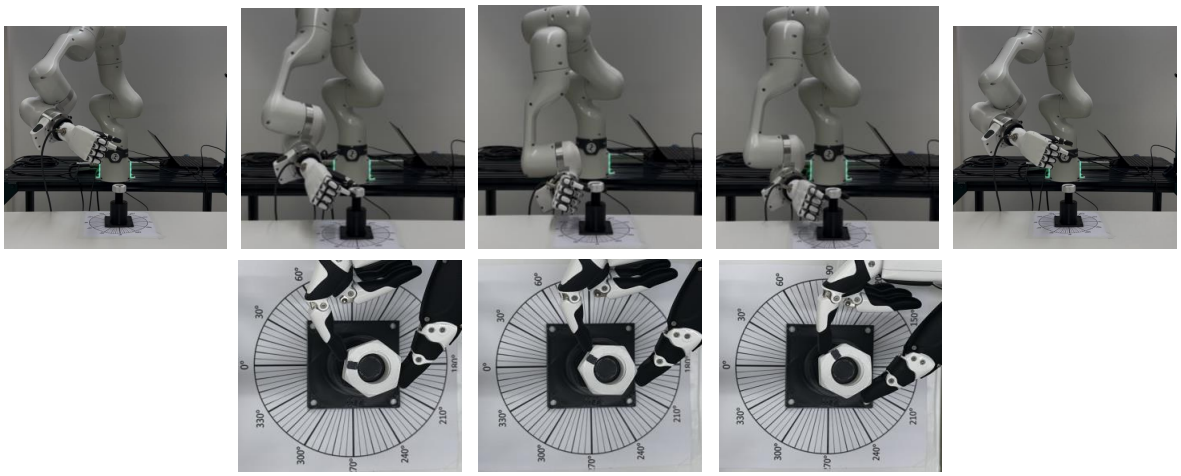


Figure 5.2: One threading cycle on the physical robot. Top, from left to right: approach, grasp, rotation, goal reaching, and release with return to the reset pose. Bottom: top-down views of the angular scale under the bolt during grasp, rotation, and goal reaching; the scale is read at the end of every goal-reaching phase to compute the cyclic and ten-cycle tracking errors.

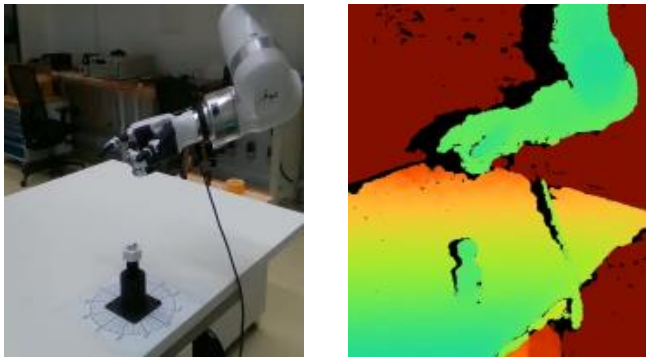


Figure 5.3: Student observation on the physical system: the RGB image (left) and the metric depth image (right) of the RealSense D415 after center-cropping and downscaling to 320×180. The depth image is back-projected through the calibrated intrinsics into the point cloud that the student encodes.

5.2 Zero-Shot Transfer Protocol

The protocol tests RQ3 directly: the student is trained on simulated data only and deployed unchanged. The student of the full method is trained for 300 epochs on the simulated dataset $\mathcal{D} = \mathcal{D}_{\text{sim}}$ of Section 3.4.5 with eight-step chunks. It is deployed on the physical robot with a prediction horizon of eight and sixteen ODE integration steps. No real-world images, trajectories, or corrections enter its training, and it is not fine-tuned on the robot. The transfer therefore rests entirely on the four properties of Section 3.4.4. The calibrated camera of Section 5.1 is the one that rendered the demonstrations. The point-cloud crop box removes the physical background before encoding. The demonstrations were rendered under lighting, texture, and background randomization. The Cartesian targets are executed by the impedance controller described above. The student receives the progress scalar from the elapsed trial time and the process phase from the physical events of the cycle, as in simulation (Section 3.5.3).

5.3 Results on the Training Nut

Table 5.1 is the hardware counterpart of the simulation ablation on the M24 hexagonal nut. It answers RQ3 on the training nut, and it checks whether the comparisons of RQ1 and RQ2 carry over to hardware. Three of the five students are taken to hardware: the full method (e), the single-cycle student (a), and the ResNet student (b). The single-cycle student tests whether the advantage of multi-cycle distillation persists on the physical system, and the ResNet student tests whether the image-space representation transfers as well as the point cloud. The two regression heads (c) and (d) are evaluated in simulation only. The hardware results are TBD.

Table 5.1: Real-world zero-shot ablations on the M24 hexagonal nut over ten trials, with the same conventions as Table 4.2. Ablations (c) and (d) are evaluated in simulation only.

Ablation	Completed cycles ↑	Cyclic error → 0	Ten-cycle error → 0
(a) single-cycle student	TBD	TBD	TBD
(b) ResNet encoder	TBD	TBD	TBD
(e) full method	TBD	TBD	TBD

5.4 Generalization across Nut Sizes and Geometries

The training nut alone does not show how far the zero-shot transfer reaches, so the full method is also evaluated on all twelve test objects. Table 5.2 reports ten trials per object: the M24, M30, and M36 hexagonal nuts, whose profile the demonstrations show, and the octagonal, rectangular, and round profiles of each size, which it has never seen. The study tests two kinds of generalization. Across sizes, the student must infer the nut size from the point cloud, since it receives no explicit size input. Across profiles, the question is whether the learned pinch-and-re-grasp strategy, which the demonstrations show only on hexagons, transfers to a different face topology. The results are TBD.

Table 5.2: Real-world zero-shot transfer across nut sizes and profiles over ten trials per nut. Values are mean ± standard deviation; the octagonal, rectangular, and round rows are out-of-distribution geometries.

Size	Profile	Completed cycles ↑	Cyclic error → 0	Ten-cycle error → 0
M24	hexagonal	TBD	TBD	TBD
	octagonal	TBD	TBD	TBD
	rectangular	TBD	TBD	TBD
	round	TBD	TBD	TBD
M30	hexagonal	TBD	TBD	TBD
	octagonal	TBD	TBD	TBD
	rectangular	TBD	TBD	TBD
	round	TBD	TBD	TBD
M36	hexagonal	TBD	TBD	TBD
	octagonal	TBD	TBD	TBD
	rectangular	TBD	TBD	TBD
	round	TBD	TBD	TBD

5.5 Long-Horizon Stress

Beyond the ten-cycle protocol, the student is run on hardware for as many cycles as it sustains, to measure how far its horizon extends beyond the ten cycles of the demonstrations. Every revolution raises the nut by one pitch. After enough cycles, the nut therefore leaves the axial range that the demonstrations cover, since the teacher that generated them was trained with up to 20 mm of initial-height randomization. The stress trials thus measure how far the learned behavior reaches along the bolt. The cycle counts per nut are TBD.

Together, the three studies provide the hardware evidence for RQ3: transfer on the training nut, generalization across sizes and profiles, and the reach of the behavior beyond the demonstrated horizon. Chapter 6 interprets this evidence alongside the simulation results of Chapter 4.

Chapter 6

Discussion

This chapter interprets the results of Chapters 4 and 5. Section 6.1 answers the three research questions of Section 1.4 as far as the available results allow. Section 6.2 explains why the deployed student is distilled offline rather than with online DAgger, and Section 6.3 draws the lesson of the action and observation contracts. Section 6.4 returns to the assembly motivation and asks how far the demonstrated skill is from industrial threading. Section 6.5 states the limitations of the work.

6.1 Interpretation per Research Question

Decomposition during data generation. All three research questions presuppose a teacher that produces complete multi-cycle demonstrations, so the teacher result comes first. PPO learns a robust threading turn once the thread is constrained kinematically and the learned task is one turn from an engaged nut. The kinematic constraint makes the learning problem well posed: it ties axial progress to rotation, so the progress reward can only be earned by turning. The teacher also answers difficulty (b) of Section 1.2.2: with the hand restricted to three commanded pinch degrees of freedom, it acquires the pinch grasp and keeps it under the turning load, and losing the grasp after it has latched ends the episode, which makes grasp retention a hard constraint (Section 3.3). Trained under initial-yaw, axial-height, and controller randomization, the retained teacher, chained into the hybrid teacher, completes ten consecutive cycles in all 32 candidate environments (Section 4.2). The RL teacher solves one turn, and the hybrid rollout extends it to as many cycles as the axial range permits. The turn also needs no force sensing: the actor of the retained teacher observes no force channel. Hardware-equivalent force observations are implemented for teacher and student and remain to be evaluated (Section 4.4.5).

RQ1: unified multi-cycle student. RQ1 tests the deployment side of difficulty (c) of Section 1.2.3. The full method completes all ten cycles in every simulated trial and lags the target by less than two degrees per cycle. The content of the data explains why. In a single-cycle demonstration the nut is always grasped at its reset height, so a student trained on it never has to read the height from the point cloud. In a multi-cycle demonstration the nut rises by one sixth of a pitch per cycle, and the student learns to track it. The global progress scalar supports the multi-cycle student, but it does not supply the axial information. That information comes from the coverage of the axial state in the demonstrations. The single-cycle ablation (a) receives the same progress scalar and therefore isolates this effect; its result is TBD. RQ1 thus tests the principle that unifying the complete cyclic behavior in one dataset,

rather than invoking a per-cycle policy repeatedly, yields a policy that sustains consecutive cycles.

RQ2: student design. RQ2 has two parts, the visual representation and the action head, and each of the ablations (b) to (d) changes one of them. On the representation side, the student does not need to localize the nut to sub-millimeter accuracy. The engagement of the nut on the bolt and the compliance of the impedance controller absorb lateral error. What the student needs is the orientation of the nut faces for the re-grasp and the height of the nut for the approach. A representation that encodes position but not orientation would let the fingers close on the right spot with the wrong face alignment. The point-cloud encoder consumes the metric geometry of nut, bolt, and hand directly. Pretrained on sweeps and fine-tuned on hybrid-teacher images, it recovers the nut pose to 1.9 mm and 3.8° on held-out teacher episodes, with about 60 000 parameters and no pretrained image backbone (Section 4.4.3). This precision suffices for ten cycles in simulation. The image-space encoder of ablation (b) also receives depth and per-pixel coordinates, so (b) tests whether the geometric structure of the representation, rather than the depth input alone, decides the closed-loop result. Random point sampling adds an augmentation that the image encoder does not receive. The closed-loop ablation (b) and the pose probe quantify the difference; their results are TBD.

On the head side, the chunked Transformer regression head (d) shares the backbone, the chunk length, and the temporal ensembling of the flow student, so the difference between (d) and (e) isolates the generative objective. Two mechanisms favor the flow head. The first is multimodality of the labels. For near-identical observations, the dataset contains the teacher's nominal action in one demonstration and its recovery after an impulse in another. A squared-error head converges to their average, an action that neither demonstration contains, whereas the flow head samples one of them. The second is the structure of the prediction. The flow head predicts a velocity from a noisy chunk toward the data, which is a denoising problem that is easier to fit than a direct regression from observation to chunk. Through the sixteen integration steps it refines all eight steps of the chunk jointly, with full attention among them. The recurrent regression head (c) additionally removes chunking. The results of (c) and (d) on the simulated dataset are TBD.

RQ3: zero-shot sim-to-real transfer. RQ3 tests the transfer side of difficulty (a) of Section 1.2.1. The student is trained on simulated data only, and its transfer rests on four properties of the pipeline (Section 3.4.4). The calibrated camera model is meant to make the simulated and real point clouds geometrically comparable. The crop box removes the background, which is where rendered and real images differ most. Lighting, texture, and background randomization, together with image-space color augmentation, are meant to cover the remaining appearance variation of the points that enter the encoder. The shared Cartesian action path executes the student's commands through a controller that reproduces the simulated control law, and the teacher was trained under randomized gains and noisy pose inputs. The same geometric property that RQ2 relies on matters here: inside the crop box the point cloud is identical in both domains up to depth noise and calibration error, whereas an image-space encoder faces the full appearance gap between rendered and real images. The approach requires no real-world data collection and no training on the robot. The hardware results that test it are TBD.

6.2 Offline versus Online Distillation

The project set out to distill the teacher online with DAgger, as the predecessor thesis did [26], and it ends with an offline flow student. The comparison in Section 4.5 shows why. Depending on who controls the environment, the online recurrent students learn the turn, the transition, or both. The evaluation ladder shows that they fit their labels and execute them correctly under teacher-forced observations. The offline flow student sustains ten cycles under its own observations, and three properties plausibly explain the difference. Its training data contain explicit recoveries from disturbances in every phase. Its chunked generative prediction with temporal ensembling smooths single-step errors. Its progress and phase conditions keep it synchronized with the physical cycle.

The advantage of DAgger, labels on student-visited states, is obtained here in a different form. The impulses move the teacher into off-nominal states and record its recovery. The online flow variant cannot provide labels on student-visited states as implemented. Its chunked labels must be produced by the teacher alone, so it runs at $\beta = 1$, obtains labels on teacher-visited states only, and reduces to online collection of the same data. The same constraint rules out the mixture schedules that the one-step recurrent student uses. An online variant on student-visited states would need multi-step teacher recovery fragments collected from student states.

The offline dataset has two further practical advantages. It is rendered once under visual randomization and shared unchanged by every student and ablation, so the comparisons of Chapter 4 differ only in the component under test. And one student trains in about thirty minutes.

6.3 Action and Observation Contracts

The project implemented two action contracts and settled on the Cartesian one. The joint-PD contract was motivated by deployment convenience and by the expectation that joint references are easier for a controller to track than Cartesian targets. Recorded joint references replayed at the physics rate do reproduce the teacher motion, with an arm-joint RMSE of 0.0037 rad (Section 3.2.3). As a learning target, however, the Cartesian contract is the stronger choice. The joint-PD contract converts the teacher's bounded Cartesian action into seventeen physical-unit references at the physics rate and asks the student to emit one such reference per 15 Hz step. This introduces a hold error that the teacher never produced. It mixes physical units with widely different scales in one regression target (Section 3.2.3). It also removes the bolt-relative structure of the action that lets the student generalize across nut heights. The Cartesian contract keeps the student in the teacher's own action space and executes it through the unchanged controller, including the action smoothing that the teacher was trained with. With the same demonstrations, it turns an equally good offline fit into closed-loop progress (Section 4.4.1). A student should therefore imitate the action the teacher produced, in the units the teacher produced it, and let the controller do the conversion it was designed for.

The observation contract follows the same logic. The student receives the previous filtered action and the controlled joints, which is what the teacher's history contains, and two low-dimensional conditions that the demonstration provides without privileged state. The global progress scalar assumes a nominally timed trial. In the ten-cycle simulation run, the student's release times drift behind those of its source demonstration by 7.4 s at the tenth cycle without affecting completion. The process-phase encoding keeps the student synchronized within each cycle, and the drift does not lead to a failed transition. For trials that run much longer

than the demonstrations, the phase should be inferred from the observation history inside the policy rather than supplied as an external signal. Chapter 7 returns to this as future work.

6.4 What This Means for Robotic Assembly

The motivation of this thesis was assembly with threaded joints (Section 1.1). Measured against an industrial fastening step, the demonstrated skill covers the run-down of an engaged nut and leaves four gaps.

The first gap is the start of the task. Every trial starts with the nut already engaged on the bolt (Section 3.2.2). In assembly the nut arrives loose and has to be seated on the thread first, which is a separate skill with a narrow entry configuration. The project has a pickup teacher that grasps and lifts a loose nut, with 99.8 % pinch success on the FR3 in simulation, and a full-task variant of the hybrid teacher that prepends it (Section 3.4.2). In the single-seed action-contract study, however, a privileged student held the engagement goal in only 2 of 10 trials (Table 4.3). Engagement is therefore the least solved part of the complete task.

The second gap is the end of the task. A trial ends after ten turns of 60°, which advance the nut by one and two thirds of a pitch, 5.0 mm on the M24 nut. The nut is never seated against a part, and it is never tightened to a preload. The simulated thread resists the turn only with a smoothed friction torque whose Coulomb term is 2×10^{-3} N m, which models a free-running nut. The actor observes no force, and the reward penalizes excess wrist force. A tightening phase would demand that the pinch transmit a much larger torque and that the policy sense when the joint is tight, which neither the thread model nor the observation contract supports.

The third gap is the grasp. The student executes one strategy, a thumb–index pinch with three commanded hand joints, learned on the face topology of a hexagon. People change the grasp as the resistance grows (Section 1.2.2), and a tightening phase would likely require such a change. The held-out octagonal, rectangular, and round profiles of Section 5.4 test how far the single strategy carries; their results are TBD.

The fourth gap is cycle time. The ten-cycle student completed its simulated trial in 1072 control ticks at 15 Hz, about 71 s, or roughly seven seconds per 60° cycle (Section 4.4.3). The phase-conditioned privileged student showed the same cycle durations of 6.7 to 7.1 s (Section 4.4.2). Release and return account for a fixed share of every cycle, since their scripted waypoints are interpolated over 0.7 s per segment and 1.0 s for the final return. Because each cycle turns the nut by 60°, a full revolution takes six cycles, which is far slower than the continuous rotation of a powered spindle.

These gaps do not erase what the result shows. In simulation, a learned policy controls a dexterous hand through repeated grasp, turn, release, and return, and it re-grasps the nut at whatever angle and height the previous cycle left it. It perceives the nut from one fixed camera, and it is trained without any real data. The same hand also picks up a loose nut in simulation. Whether this versatility outweighs the speed of a dedicated tool depends on the application, and the remaining gaps define the work that separates the two (Chapter 7).

6.5 Limitations

Section 6.4 described the task-level gaps to industrial threading; this section states the limitations of the method and of the evidence. The demonstrated behavior is a pinch-grasp re-grasp strategy, and the student has no mechanism to discover a different grasp for a profile that re-

quires one. The axial range of the demonstrations is bounded by the 20 mm initial-height randomization of the teacher, and the global progress clock is defined over the nominal trial length. The teacher's thread model is a kinematic simplification. In simulation the nut therefore cannot jam or cross-thread, and the thread resists only through a small friction torque on the revolute joint. The randomization ranges that are meant to cover the resulting sim-to-real gap have not yet been fitted to measured statistics of the physical sensor. The camera calibration has a translation fit RMSE of 27 mm and a rotation fit RMSE of 3.5° (Section 5.1); it is measured but not deployment-validated, and the dependence of the student on calibration error has not been quantified. The evaluation uses ten trials per condition and reports means and standard deviations without confidence intervals. Several design ablations in Chapter 4 are matched single rollouts at one seed, which establish the direction of an effect but not its variance. Chapter 7 summarizes what the thesis achieved and turns these gaps and limitations into future work.

Chapter 7

Conclusion and Future Work

This chapter summarizes what the thesis achieved, answers the research questions, and outlines the work that remains between the demonstrated skill and robotic assembly with threaded joints.

7.1 Conclusion

Threaded joints can be released again and their threads are standardized [1], and fastening a nut is still largely left to people and specialized tools. This thesis asked how a dexterous robot can learn to thread a nut. Threading poses three difficulties at once: thread contact is hard to simulate, so the skill has to be learned in simulation and transferred; a dexterous hand has a large action space and many ways to solve the task; and threading is a long-horizon cyclic task in which errors compound across cycles and phase transitions.

The thesis answers these difficulties with one principle: decompose during data generation, unify at deployment. During data generation the cycle is decomposed into a turning skill and scripted transitions. A privileged asymmetric actor–critic learns the pinch-grasp turn in a simplified thread simulation in which axial progress follows from rotation. Scripted release and return chain the turns into a hybrid teacher that produces complete multi-cycle demonstrations with physically gated phase boundaries. At deployment the behavior is unified again. One visuomotor student, distilled from the demonstrations by conditional flow matching, executes the whole cycle from an RGB-D camera and proprioception. It is trained on simulated data only and carried to hardware by a calibrated camera model, a metric point cloud cropped to the workspace, visual domain randomization, and a shared Cartesian action path.

The evidence supports the principle in simulation. Chained into the hybrid teacher, the RL teacher completed ten consecutive cycles in all 32 candidate environments. The distilled student completes all ten cycles in every trial on the M24 hexagonal nut, with a mean cyclic tracking error of $-1.77°$ and a ten-cycle error of $-17.7°$. The single-seed design studies show why each component of the student is present. The Cartesian action contract turns an equally good offline fit into closed-loop progress. The progress and phase conditions keep the student synchronized with the physical cycle. Demonstrations with recoveries in every phase plausibly let it correct its own deviations, which is where the online recurrent students fell behind.

The three research questions are answered as follows.

RQ1 One student distilled from complete multi-cycle demonstrations sustains all ten cycles in simulation, and its comparison with a single-cycle policy invoked repeatedly is TBD.

RQ2 The flow-matching student with a point-cloud encoder sustains the cyclic task, and its comparison with deterministic LSTM and Transformer regression and with an image-space encoder is TBD.

RQ3 The student is trained on simulated data only and deployed without real data or training on the robot, and the result of its zero-shot transfer on the M24, M30, and M36 nuts and on held-out profiles is TBD.

Returning to the motivation of Section 1.1: in simulation, a dexterous hand can now thread a nut over many consecutive cycles with a learned policy, re-grasping it in every cycle from camera data at whatever angle and height the previous cycle left it, and the policy is trained without real data. Whether it does so on the physical system is TBD (RQ3). What it cannot yet do is start from a loose nut, tighten the joint, choose a different grasp when the resistance grows, or match the speed of a dedicated tool (Section 6.4). These gaps define the future work.

7.2 Future Work

The first extension is autonomous engagement of a loose nut. The pickup teacher already grasps, lifts, and transports the nut, and the full-task variant of the hybrid teacher chains it with threading. Seating the nut on the bolt is a separate skill with a narrow entry configuration. It will most likely require fingertip-force and tactile feedback, and the hardware-equivalent fingertip-force and joint-torque observation channels that are implemented for teacher and student are the starting point.

The second extension is tightening to a preload. A threading cycle that ends with the nut tight needs a thread model whose resistance rises when the nut seats, a grasp that transmits the resulting torque, and a policy that senses when the joint is tight. The same force channels are the natural inputs for such a policy.

The third extension is a longer horizon. The axial range of the demonstrations is bounded by the 20 mm initial-height randomization of the teacher. A curriculum that widens this range toward the full usable thread length would extend the horizon of the teacher and of the student with it. The global progress clock should be replaced at the same time, because it binds the student to a nominal trial duration. Inferring the phase from the observation history inside the policy would remove that dependency.

The fourth extension concerns the transfer. The randomization ranges of the simulated camera should be fitted to measured statistics of the physical D415, including depth failures, exposure, motion blur, and camera and control latency. The depth-corruption model that the encoder pretraining already provides can then be applied to the student's training data as well.

The fifth extension concerns the grasp. Training on more than one face topology would extend the re-grasp strategy to profiles that a hexagon does not cover, and other grasp topologies than the thumb–index pinch would let the hand adapt its grasp to the resistance of the thread.

The last extension completes the distillation study. The online flow-matching student is implemented and remains to be evaluated. Collecting multi-step teacher recovery fragments from states that the student visits would give it on-policy labels without violating the constraint that a chunk label must be produced by the teacher alone. The pending sweeps over the action horizon, the number of integration steps, and the temporal-ensemble decay, and the closed-loop comparison of the point-cloud encoder with the Adapt3R-style encoder, would complete the design study.

Appendix A

Hyperparameters and Interfaces

A.1 Teacher Observations

Table A.1: Actor and critic observations of the single-size M24 threading teacher. Noise is applied to the actor only. The multi-size teacher appends a clean 3-D one-hot nut-size code to both.

Input	Component	Dim	Noise (actor)
actor (39-D)	fingertip position relative to bolt frame	3	$\mathcal{N}(0, 0.25\,\text{mm})$ per axis; bolt frame $\mathcal{N}(0, 1\,\text{mm})$ per reset
	fingertip quaternion (WXYZ)	4	random-axis angle $\mathcal{N}(0, 0.1°)$
	end-effector linear velocity	3	finite difference of noisy position
	end-effector angular velocity	3	finite difference of noisy quaternion
	fingertip position relative to held nut	3	$\mathcal{N}(0, 0.25\,\text{mm})$ per axis, each body
	held-nut quaternion (WXYZ)	4	random-axis angle $\mathcal{N}(0, 0.1°)$
	target position error	3	inherits nut noise
	target quaternion error (WXYZ)	4	inherits nut noise
	grasp-phase flag	1	none
	thumb / index flexion force	2	$\mathcal{N}(0, 1\,\text{N})$, clamped to ≥ 0
	previous filtered action	9	none
critic (69-D)	fingertip position, quaternion, velocities	13	clean
	arm joint positions	7	clean
	held-nut position, position relative to bolt, quaternion	10	clean
	bolt position and quaternion	7	clean
	controller proportional gains	6	sampled per reset
	action smoothing factor	1	sampled per reset
	thumb / index flexion force	2	clean
	position and rotation action thresholds	6	sampled per reset
	target position, quaternion, progress	8	clean
	previous filtered action	9	none

A.2 Teacher Hyperparameters

A.3 Student Hyperparameters

A.4 Dataset Summary

Table A.2: PPO hyperparameters of the privileged teacher (rl_games, asymmetric actor–critic).

Parameter	Value
parallel environments	512
epochs	200
rollout horizon	256 steps
minibatch size	512
mini-epochs per update	4
recurrent sequence length	128
discount γ / GAE τ	0.995 / 0.95
learning rate	10^{-4}, adaptive with KL threshold 0.008
clipping ϵ	0.2
critic coefficient	2
entropy coefficient	0
gradient-norm clip	1.0
bounds loss coefficient	10^{-4}
actor / critic network	LSTM 2×1024 (layer norm) $\rightarrow$ MLP 512–128–64, ELU
action distribution	diagonal Gaussian, state-dependent σ
mixed precision	yes

Table A.3: Hyperparameters of the conditional flow-matching student.

Parameter	Value
action contract	9-D Cartesian, $[-1, 1]$
action horizon H	8 (0.53 s at 15 Hz)
target offset	0 (pre-step alignment)
terminal-hold padding	yes
proprioception	29-D (10 q, 10 $\dot{q}$, previous action)
proprioception encoder	MLP 128–128, ELU
visual feature	64-D (DP3: 4096 XYZRGB points, PointNet 64–128–256, max pool)
condition width	256
progress / phase embeddings	MLP 1–256–256 / 3–256–256, ELU
flow-time embedding	sinusoidal, 64-D
velocity network	Transformer, 4 layers, 8 heads, width 512, FF 2048, GELU, pre-norm
probability path	conditional optimal transport (linear interpolant)
optimizer	Adam, learning rate 10^{-4}, batch 64, 300 epochs
parameter EMA decay	0.999
auxiliary loss	nut position from visual feature, weight 1.0
RGB augmentation	saturation/contrast/brightness $[0.5, 1.5]$, hue $[-0.15, 0.15]$, blur $p = 0.1$
sampler	Euler, $N = 16$ steps (online training: 8)
temporal-ensemble decay	0.5

Table A.4: Demonstration data used for the reported students.

Property	Value
simulated demonstrations	20 successful ten-cycle episodes from 32 candidates
frames (15 Hz)	19 486
impulses per episode	30 (one per policy, waypoint, and return segment), 0.05–0.25 N s
visual randomization	dome light, table, nut/bolt, floor textures, background
image size	320×180 RGB and metric depth (from 640×480)
DP3 points per frame	10 542 min / 12 374 median / 15 366 max valid; 4096 sampled

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